IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier xxx

Beamforming Designs for Enabling Symbiotic BackCom Multiple Access Under Imperfect CSI

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ABSTRACT This paper investigates the optimization of the primary base station (BS) to effectively support a symbiotic radio (SR) network between a primary network consisting of a single user and the secondary network comprising multiple tags of a backscatter network. The tags use non-orthogonal multiple access for communication, and we consider passive tags and reconfigurable tags. To decode data successfully, the primary user adopts a successive interference cancellation technique. However, in practical networks, imperfect channel state information is inevitable and can significantly degrade network performance. Surprisingly, this issue has not been previously investigated. Thus, considering these factors, we aim to optimize the beamforming vectors at the BS, supporting both the primary and backscatter networks, while simultaneously maximizing the weighted sum rate or minimizing transmit power. These optimization problems are non-convex, and we develop a combination of alternative optimization, fractional programming, and semi-definite relaxation techniques for solutions. We conduct evaluations of the proposed algorithms considering both fixed and reconfigurable tags with variable reflection coefficients. Furthermore, extensive numerical simulations are presented to demonstrate the validity and performance gains achieved by the proposed schemes. The results of our study highlight the improved benefits that SR systems can offer through the utilization of our optimization schemes.

INDEX TERMS Backscatter communication, Symbiotic radio, Non-orthogonal multiple access.

I. INTRODUCTION

A. BACKGROUND AND SCOPE

AMBIENT power-enabled (i.e., battery-free) Internet-of-Things (IoT) is becoming a highly active research topic, and the 3rd generation partnership project (3GPP) has opened a new study item [1], [2]. Ambient IoT networks refer to a network of interconnected devices that can sense, collect, and share data about their environment without requiring explicit human input or control. These devices are equipped with sensors, microprocessors, and wireless connectivity, allowing them to communicate with each other and exchange information in real time. Ambient IoT networks can be used in a variety of settings, such as smart homes, smart cities, industrial IoT, and healthcare. In a healthcare setting, for example, ambient IoT devices can monitor patients' vital signs, track their movements, and alert healthcare providers to potential health issues. Symbiotic radio (SR) networks [3], [4] are one of the means to enable and support ambient IoT networks. Before delving into that, we must discuss some of the basics of backscatter IoT networks.

Reductions in IoT device size, complexity, and power consumption can facilitate the deployment of tens or even hundreds of billions of IoT devices for many applications, including logistics, inventory management, warehousing, manufacturing, and many more [5]. For example, consider a use case (U1) where a mobile user enters a large shopping mall with its own IoT network and the commercial Wi-Fi network (Fig. 1). The user can immediately access information about nearby facilities and shops, or the network can navigate the user to a certain shop/place based on the user's request, which existing fourth-generation (4G) and fifth-generation (5G) positioning cannot do. Such networks can be inexpensively

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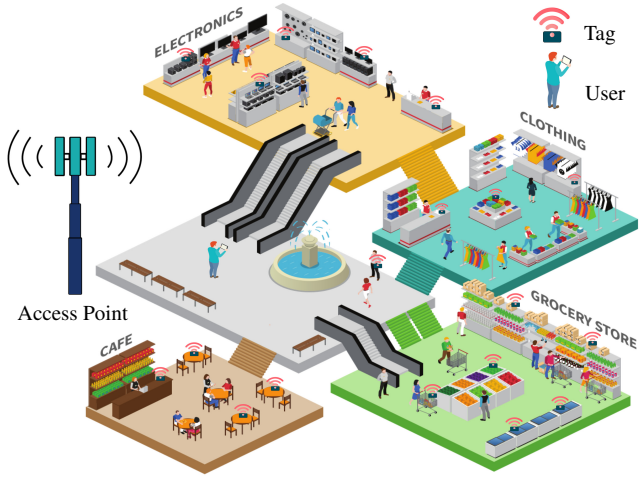

FIGURE 1: A shopping mall use case.

created by using large numbers of simple IoT devices.

B. BACKSCATTER NETWORKS

However, IoT devices are powered with batteries that need to be replaced or recharged manually. This incurs significant maintenance costs, substantial environmental concerns, and even safety hazards for specific use cases (e.g., wireless sensors for power and petroleum industries). On the other hand, since the output power of energy harvesting (EH) typically ranges from $1\,\mu\mathrm{W}$ to a few hundred $\mu\mathrm{W}$, existing cellular devices with $10\,\mathrm{mW}$ peak power consumption may not operate well with EH [2], [5]. Hence, as a remedy, batteryless devices with no energy storage capabilities, i.e., passive devices, or devices with limited energy storage, i.e., semi-passive devices, will facilitate massive connectivity expectations [6]. In particular, ambient backscatter communication (AmBC) systems may enable future ambient IoT networks and applications [3], [4], [7], [8].

These systems leverage ambient radio frequency (RF) sources, e.g., cellular base stations (BSs), television (TV) towers, and Wi-Fi access points (APs). The tags use those signals to harvest energy and backscatter data [9]–[12]. They thus do not generate the RF signals themselves. They are thus inexpensive and have ultra-low energy requirements (a few nW–$\mu\mathrm{W}$) [9]. As AmBC systems eschew new frequency spectrum allocation, the spectrum utilization improves. However, legacy signals can directly interfere, limiting the AmBC performance. Sophisticated methods are thus required to deal with direct link interference [13], [14].

Passive tags can be classified into two types: those with fixed reflection coefficients and those with reconfigurable tags that have variable reflection coefficients. Fixed tags are cost-effective and provide a practical networking solution due to their small, thin, and low-cost design. On the other hand, reconfigurable tags, though more expensive, have the potential to enhance network performance. However, because these tags need to dynamically adjust their reflection coefficients, a more complicated tag architecture and higher power consumption are necessary.

Despite their complexity, reconfigurable tags have been studied extensively [15]–[17]. However, passive tags with fixed reflection coefficients have the advantage of simplified network design and maintenance, as shown in recent studies [18], [19]. Therefore, while reconfigurable tags may offer benefits in certain applications, it is essential to consider the advantages and disadvantages of both types of tags when designing a passive tag system.

C. SYMBIOTIC RADIO NETWORKS

An SR network is the coexistence of two networks (primary and secondary) together cooperatively to share information and resources over a shared radio frequency spectrum [3], [4]. The nodes thus actively collaborate with each other to optimize network performance, reduce interference, and conserve power. The concept of SR networking draws inspiration from the biological concept of symbiosis, where two or more species coexist in close proximity and derive mutual benefits from each other. Symbiotic relationships are characterized by the exchange of resources such as food, shelter, and protection, among others, leading to enhanced persistence and expansion of ecological niches. Positive species interactions are crucial in maintaining diverse ecosystems and have been shown to increase the resilience of organisms in many ecosystems. For example, the diversity of mycorrhizal symbionts is positively correlated with the biodiversity of plant communities, resulting in greater stability and longer persistence of the community [20].

The most common types of biological symbiosis include:

1) **Mutualism:** Both species involved in the relationship benefit from each other. For example, some species of plants and insects have mutualistic relationships where the insects pollinate the plants and the plants provide food for the insects. For example, certain fungi form mutualistic symbiotic associations with terrestrial plant species to maintain plant biodiversity, ecosystem variability, and productivity [21].
2) **Commensalism:** One species benefits from the relationship while the other is neither helped nor harmed. For example, some species of birds build their nests in trees, which provides them with shelter, but does not harm the tree.
3) **Parasitism:** One species benefits at the expense of the other. For example, ticks and fleas are parasites that feed on the blood of their hosts, causing harm or even death.

These concepts can be adapted for wireless networks readily. For instance, in an SR network, the nodes in primary and secondary networks cooperate to share their resources, such as bandwidth and power, and coordinate their transmissions to avoid interference and improve network performance. For the SR network considered in Fig. 2, the secondary consists of multiple tags that talk to the primary user. Thus, the user must employ a common receiver to decode both tag

and primary signals [3], [8], [22]. The user first decodes the primary signal and then subtracts it from the received signal to decode the tag signals. However, depending on the primary and tag data rates, the symbiosis could be either commensalism or parasitism. In commensal SR (CSR), the tag data rate is substantially lower than the primary one, and the tag signal can be treated as a multi-path component [3]. On the other hand, in parasitic SR (PSR) the primary and tag data rates are equal; tag signals directly interfere with primary signals. In either case, multi-objective optimization techniques are necessary to maintain the performance of both the primary and the tags [7], [15].

While cognitive radio also refers to the coexistence of primary and secondary networks, SR differs from traditional cognitive radio networks, where the only shared resource is bandwidth. The primary goal of cognitive radio is to improve the utilization of the radio frequency spectrum by identifying and utilizing unused or underutilized frequency bands while avoiding interference with licensed users [23], [24]. The secondary network has a separate receiver and the primary network has no collaboration with the secondary one. On the other hand, SR focuses on collaboration and coordination among the two networks to optimize the overall system performance in terms of capacity, energy efficiency, and reliability.

An SR comprises a primary network of $N \geq 1$ users and a secondary network of $K \geq 1$ backscatter tags. The basic SR setup with $N = K = 1$ is widely investigated [3]. When $N > 1$ and $K = 1$, it enables multiple access in the primary network while only serving a single tag in the secondary network [25], [26]. References [25], [26] propose optimization frameworks with different goals, i.e, weight sum rate maximization and transmit power minimization in [25] and primary energy efficiency maximization in [26]. Reference [17] investigates the general system setup, i.e., $N > 1$ and $K > 1$, with a full-duplex (FD) BS that decodes the tags data. It specifically develops a tag throughput maximization scheme by optimizing tags' reflection coefficients. On the other hand, this paper investigates the case with $N = 1$ and $K > 1$ [15], [16]. This setup enables multiple access for the secondary (or backscatter) network.

Next, we present the related works to provide the context and background for our work.

D. PREVIOUS WORKS ON SR WITH NOMA TAGS

Because passive tags have limited processing capability and power, how they can use SR and non-orthogonal multiple access (NOMA) has not received much interest, except for [15]–[17]. In contrast to orthogonal multiple access (OMA), NOMA allows multiple users to share the same orthogonal channel simultaneously. In power-domain NOMA, multiple user signals are distinguished based on the channel gain differences, and individual user signal detection is achieved using successive interference cancellation (SIC) [27]. Thus, supporting multiple tags with NOMA and SR requires further research. Below, we briefly describe these contributions.

In [17], an FD BS serves downlink users through space division multiple access (SDMA) and simultaneously decodes multiple tag signals. The AP uses SIC decoding for this purpose. This work finds the tag reflection coefficient to maximize the uplink throughput while keeping the interference experienced by downlink users lower than a threshold.

Reference [15] investigates a NOMA-assisted dynamic time-division-multiple-access (TDMA) SR scheme. Tags are divided into several clusters, and only the tags in one cluster transmit during each time slot via NOMA. Under primary user rate constraint and tag EH constraints, the minimum throughput among all tags is maximized by jointly optimizing tag reflection coefficients, the BS power allocation in different time slots, and the time allocation. The NOMA+TDMA scheme is further investigated in [16] for an FD SR network, where an AP transmits downlink orthogonal frequency division multiplexing signals to a primary user and simultaneously receives signals from multiple tags. The authors then maximize the minimum throughput among all tags by optimizing the AP sub-carrier power allocation, time allocation, and reflection coefficients, subject to the primary user throughput requirement and the tag EH constraints.

Closely related research works [15], [16] consider single-antenna BS and perform the resource allocation under perfect channel state information (CSI). Besides, the effort is to ensure fairness between the tags while meeting the rate requirement of the primary user. Additionally, [15], [16], and [17] all consider optimizing the tag's reflection coefficient, i.e., reconfigurable tags, which may not be feasible with passive tags because of the limited power availability. Table 1 compares and contrasts our work with the related literature.

On the other hand, symbiosis of a NOMA-enabled primary network and a single tag, i.e., $N > 1$ and $K = 1$, have been investigated in [19], [25], [26], [28]–[33]. In particular, in [28], a single-antenna BS serves two NOMA primary users and a tag. The rate of the primary network is then maximized under SIC decoding error by jointly optimizing users' power allocation factors and the reflection coefficient. Reference [26] also investigates the primary system's energy efficiency by extending [28] to a multi-cell scenario. References [30] and [33] evaluate the outage probability of the primary users and tag. Reference [32] considers a RIS-assisted NOMA system with a tag. Here, the RIS assists the primary cell-edge user's transmission assuming direct communication with the transmitter is unavailable. The overall energy efficiency of the system is then maximized by jointly optimizing the RIS phase shifts and the power allocation for the NOMA users, considering their rates.

Additionally, our previous work [25] investigates the BS transmit beamforming design and the power allocation to the NOMA users under SIC decoding errors. The CSI knowledge is assumed to be perfect and the tag uses a fixed reflection coefficient. Whereas, this paper enables multiple access for backscatter tags along with a single user (i.e., $N = 1$ and $K > 1$) under imperfect CSI. We also quantify the effects of fixed reflection coefficients (i.e., passive tags) and variable

reflection coefficients at the tags (i.e., reconfigurable tags).

E. PROBLEM STATEMENT AND CONTRIBUTIONS

We consider a prototype symbiotic network of a single user and multiple tags (Fig. 2). The tags utilize the primary BS signal opportunistically to communicate with the primary user by using NOMA [34]. We will consider two types of tags: (a) passive tags with fixed reflection coefficients and (b) reconfigurable tags with variable reflection coefficients. To achieve symbiotic gains, the BS must support both the primary user and the tags. Specifically, the BS needs to shape its transmit signal, also known as beamforming, to simultaneously meet the rate requirements of the primary user and the tags, as well as the EH constraints of the tags. The ability to inexpensively connect a large number of tags to the network will open up many potential applications (e.g., Fig. 1). Moreover, symbiotic networks of primary users and tags can yield spectral and energy efficiency gains [15], [16]. Considering these goals, this study aims to develop algorithms to optimize the BS and the tag reflections of the considered network (Fig. 2).

Additionally, from a quality-of-service standpoint, it becomes essential to ensure minimum rate targets for both users and tags to guarantee a satisfactory user experience and reliable data transmission. Furthermore, energy efficiency increase by reducing the power consumption of BS units plays a pivotal role in achieving sustainable and eco-friendly operations for SR networks. To investigate all these factors, our study addresses the following significant gaps in the literature:

1) *Power consumption & massive connectivity*: Energy-efficient (green) BS designs play a crucial role in enabling sustainable ambient IoT and SR networks. Our primary objective is to minimize the total power consumption attributed to the BS, as it directly impacts the overall energy efficiency of the network. Surprisingly, previous related research [15]–[17] has not adequately addressed the critical aspect of BS power minimization or provided a quantification of the potential gains achievable through such optimizations.
2) *Primary & secondary performance*: The coexistence of primary users and tags in an SR is characterized by distinct device characteristics, such as varying bandwidth, power levels, and other factors, leading to the need for different data rates for each entity. To facilitate the symbiosis of both networks, a flexible assignment approach that prioritizes weighted sum rate maximization becomes essential. While previous studies [15]–[17] have primarily concentrated on improving the performance of secondary users, it is equally crucial to address the optimization and synergy of both primary and secondary networks.
3) *CSI estimation error*: Accurate CSI acquisition is challenging (see [35]–[39] and references therein), especially with multiple tags. However, almost all the relevant works assume perfect CSI estimation. Effects of imperfect CSI on resource allocation are thus critical to gain insight into the performance under real-world conditions. Developing robust beamforming and communication strategies can mitigate the effects of imperfect CSI and lead to more resilient and dependable SR networks.

Thus, the main objectives of this paper are to remedy these gaps and to develop algorithms to optimize the performance of the SR networks.

The main contributions of this paper can be summarized as follows:

1) We first begin with the passive tags with fixed reflection coefficients. For these, we formulate (i) maximization of the weighted sum of the primary user and the tags rates, and (ii) minimization of the BS transmit power.
   a) To achieve the first objective, we develop an algorithm to optimize the BS transmit beamforming, subject to minimum rate targets for the user and tags, the EH constraints at the tags, and preset levels of the BS transmit power and CSI estimation errors (see $\mathbf{P}_{\mathrm{w}}$ (19)). The resulting problem has a non-convex objective function and constraints. Therefore, we transform the non-convex objective function using fractional programming (FP), creating a sequence of convex problems [40]–[42]. By means of this approach, we develop a solution algorithm for this problem.
   b) For the second case, we design an algorithm to minimize the BS power by optimizing BS beamforming under the rate targets, the tag EH constraints, and a preset level of CSI estimation errors (see $\mathbf{P}_{\mathrm{p}}$ (26)). This problem is also non-convex because of the non-convex constraints. We thus utilize the semi-definite relaxation (SDR) technique [43] for our algorithmic development.
2) We next consider reconfigurable tags with variable reflection coefficients. We thus extend the weighted sum rate maximization algorithms developed above and optimize the reflection coefficients. The resultant problem is highly complicated (non-convex) because the optimization variables are coupled. Therefore, we decouple it into two sub-problems using the alternating optimization (AO) technique [44]. The sub-problems are then solved by using the FP technique [40]–[42].
3) We show that the proposed algorithms offer impressive symbiotic gains between the primary and secondary networks. Specifically, the number of connected tags increases while the BS beamforming designs satisfy both the rate requirements and the EH constraints. We also show that the proposed algorithms are robust against CSI errors.
4) We also show that, as compared to passive tags, reconfigurable tags can improve system performance at the

TABLE 1: Summary of Related Works.

Reference	Objective	Constraints			Variables†	Methodology	Imperfect CSI
		Primary rate	Tag rate	EH			
[15]	Tags minimum throughput maximization	✓	✗	✓	$\rho, \boldsymbol{\alpha}, \tau$	BCD & SCO‡	✗
[16]	Tags minimum throughput maximization	✓	✗	✓	$\rho, \boldsymbol{\alpha}, \tau$	BCD & SCO	✗
[17]	Tags throughput maximization	✗	✗	✗	$\boldsymbol{\alpha}$	–	✗
[25]§	Weighted sum rate maximization	✓	✓	✓	$\mathbf{w}, \rho$	FP	✗
	Transmit power minimization	✓	✓	✓	$p_t, \mathbf{w}, \rho$	SDR	✗
This Paper	Weighted sum rate maximization	✓	✓	✓	$\mathbf{w}\ (\boldsymbol{\alpha})$	FP	✓
	Transmit power minimization	✓	✓	✓	$p_t, \mathbf{w}\ (\boldsymbol{\alpha})$	SDR	✓

† The variables p_t, $\mathbf{w}$, ρ, α, and τ denote transmit power, transmit beamforming, power allocation factor, reflection coefficient, and time allocation.
‡ Block coordinated descent & successive convex optimization techniques.
§ Multi-user single tag system, i.e., $N > 1$ and $K = 1$.

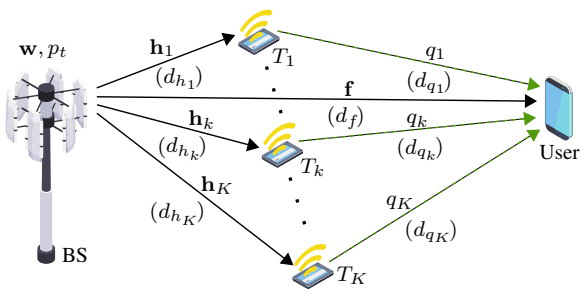

FIGURE 2: A NOMA-enabled BackCom system setup.

expense of tag architecture complexity.

5) We present extensive numerical examples to evaluate the performance of the considered network through the proposed solutions.

Notation: Lower-case bold and upper-case bold denote vectors and matrices. $\mathbf{I}_n$ is the $n \times n$ identity matrix. $\mathbf{A}^{\mathrm{T}}$, $\mathbf{A}^{\mathrm{H}}$, $\mathrm{Tr}(\mathbf{A})$, and $[\mathbf{A}]_{(m,n)}$ denote transpose, Hermitian transpose, trace, and the (m, n)-th element of $\mathbf{A}$. $\mathbb{E}\{x\}$ is the mean of x. Finally, $\mathcal{CN}(\boldsymbol{\mu}, \mathbf{R})$ is a complex Gaussian vector with mean $\boldsymbol{\mu}$ and co-variance $\mathbf{R}$. Finally, $\mathcal{K} \triangleq \{1, \ldots, K\}$, $\mathcal{K}_k \triangleq \{1, \ldots, k-1\}$, $\mathcal{K}'_k \triangleq \{k+1, \ldots, K\}$, and $\mathcal{K}_0 \triangleq \{0, 1, \ldots, K\}$.

II. SYSTEM MODEL AND PRELIMINARIES

Here, we describe the system model, channel model, and transmission model in detail.

A. SYSTEM AND CHANNEL MODELS

In Fig. 2, we consider an SR network that consists of a BS, as the primary transmitter, communicating with a primary receiver (user), and $K \geq 1$ tags, denoted by $T_k, k \in \mathcal{K}$, which leverage the primary signal to transmit their data to the user. The BS has $M \geq 1$ antennas, while the user and the tags are single-antenna devices. We assume that the tags are passive without any RF signal generation; they harvest energy from the primary signal and also reflect it to communicate their data. Therefore, the BS applies beamforming to transmit signal to the user, while powering the tags to send their information to the user. We assume the tags transmit their data via the power-domain NOMA technique [16], [34].

We consider block flat-fading channel models for the system. Thus, during each fading block, the direct-link channel from the BS to the user is denoted by $\mathbf{f} \in \mathbb{C}^{M \times 1}$. The backscatter-link channel through T_k is $\mathbf{g}_k = \mathbf{h}_k q_k, k \in \mathcal{K}$, which is the multiplication of the forward-link channel from the BS to T_k, denoted by $\mathbf{h}_k \in \mathbb{C}^{M \times 1}$, and the backscattered-link channel from T_k to the user, denoted by $q_k \in \mathbb{C}$. All channels are assumed to be independent quasi-static Rayleigh fading which remains constant during the coherence interval. A unified representation of all five channels is given as

$$\mathbf{a} \sim \mathcal{CN}\left(\mathbf{0}, \zeta_{\mathbf{a}} \mathbf{I}_M\right), \tag{1}$$

where $\mathbf{a} \in \{\mathbf{f}, \mathbf{h}_k, q_k\}$ and $\zeta_{\mathbf{a}}$ accounts for the large-scale path-loss and shadowing.

Channel estimation for backscatter networks poses two unique sets of challenges compared to channel estimation for conventional networks:

1) Estimations are needed for both the direct-link channel ($\mathbf{f}$) and the backscatter-link channels ($\mathbf{g}_k$ for $k \in \mathcal{K}$). The latter channels are dyadic channels, which suffer double path losses [11], [12].
2) Tags are passive radios with low power and processing capabilities and cannot transmit pilots endogenously. However, they can only reflect pilots sent by another source.

As a result, the channel estimation problem is radically different. Nevertheless, several pilot-based, blind, semi-blind, and learning-based techniques have already been developed [35], [45]–[48]. In particular, many pilot-based techniques restrict [35], [45], [46] one tag to reflect pilots while all other tags remain silent, whereas machine learning techniques are also appearing [47].

On the other hand, the recently developed time-spread pilot-based approach [48] enables simultaneous estimation of direct and multi-tag channels by carefully designing a set of orthogonal pilot sequences. We omit the details of these channel estimation techniques for the sake of brevity and refer interested readers to [35], [45]–[48].

We thus assume that the SR system (Fig. 2) operates in time-division-duplexing (TDD) mode for channel estimation and data transmission. The CSI acquisition of the SR system can be accomplished through some of the techniques mentioned above.

1) CSI Error Model

To account for the unavoidable channel estimation error, we model the true channels as

$$\mathbf{a} = \hat{\mathbf{a}} + \tilde{\mathbf{a}}, \tag{2}$$

where $\hat{\mathbf{a}} \sim \mathcal{CN}(\mathbf{0}, \delta\zeta_{\mathbf{a}}\mathbf{I}_M)$ is the channel estimate and δ, $0 \leq \delta \leq 1$, accounts for the quality of channel estimation [49], [50]. Specifically, $\delta = 0$ represents the fully imperfect or no CSI case whereas $\delta = 1$ denotes the perfect CSI case. Moreover, $\tilde{\mathbf{a}}$ represents the channel estimation error, and it is assumed that the channel estimate, $\hat{\mathbf{a}}$, and the channel estimation error, $\tilde{\mathbf{a}}$, are mutually uncorrelated. This assumption indeed holds for minimum mean-square error estimation [49], [50]. Thus, the channel estimation error can be modeled as $\tilde{\mathbf{a}} = \mathbf{a} - \hat{\mathbf{a}} \sim \mathcal{CN}(\mathbf{0}, (1-\delta)\zeta_{\mathbf{a}}\mathbf{I}_M)$ [49], [50].

B. TAG'S DATA TRANSMISSION AND EH

As mentioned before, tags are passive devices that do not require batteries or any other external power sources and rely completely on EH to operate their essential circuits and processing. However, tags must also communicate their data to the reader. Thus, data transmission and EH tasks can be done in two ways: (i) time-switching, i.e., the two tasks occur in separate time slots, and (ii) power-splitting, i.e., both tasks occur concurrently, but each consumes a fraction of the incident RF power [51]. However, time-switching requires the tags to harvest and store their energy. Passive tags may not be able to do so. Thus, we consider power-splitting mode at the tags [52].

The power-splitting operation at T_k with reflection coefficient α_k $(k \in \mathcal{K})$ can be described as follows: Suppose the incident RF power is p_k^a. Then, $\alpha_k p_k^a$ is used for data transmission and the remainder $p_k^{\text{in}} = (1-\alpha_k)p_k^a$ is used for EH. The reflection coefficient satisfies $0 < \alpha_k < 1$.

Here is the general model of an EH process of a tag. The EH circuit of the tag converts the incident RF power, p_{in}, to direct current (DC) power. In general, the amount of harvested power, p_h, at the tag can be characterized using linear or nonlinear EH models as

$$p_h = \begin{cases} \eta_b p_{\text{in}}, & \text{Linear}, \\ \Phi(p_{\text{in}}), & \text{Nonlinear}, \end{cases} \tag{3}$$

where $0 < \eta_b \leq 1$ represents the energy conversion efficiency of the linear EH model, and it is assumed to be a constant independent of the incident RF power. The nonlinear EH model, $\Phi(x)$, is defined as [53]

$$\Phi(x) = \left[\frac{p_{\max}}{\exp(-bp_0+v)}\left(\frac{1+\exp(-bp_0+v)}{1+\exp(-bx+v)} - 1\right)\right]^+, \tag{4}$$

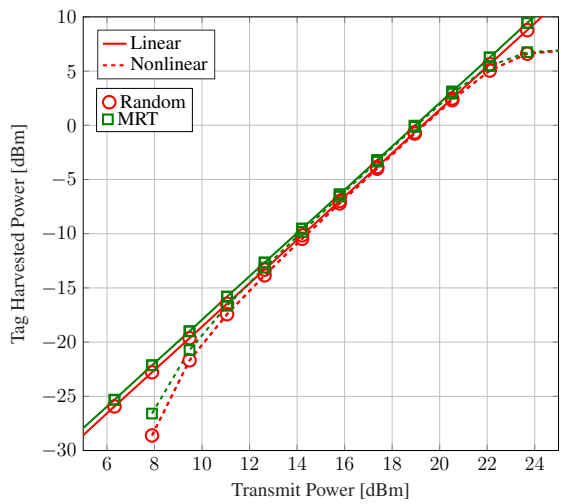

FIGURE 3: The tag's harvested power as a function of the BS transmit power with random and MRT beamforming for $M = 32$, $d_{h_1} = 5$ m, $\alpha = 0.6$, and $\eta_b = 0.6$.

where $p_{\max}$ denotes the maximum harvested power, p_0 accounts for the harvester's sensitivity threshold, and the parameters b and v control the steepness of the function $\Phi(\cdot)$. Besides, $[x]^+ = \max(x, 0)$.

Remark 1. *The nonlinear EH model in* (4) *accurately represents the nonlinear properties of practical EH circuits including a saturation plateau for large input powers and zero output for input power less than the sensitivity threshold, (e.g.,* $-20\,\text{dBm}$ *for commercially available passive radio frequency identification (RFID) tags [15], [54]) [53], [55]. In contrast, the linear EH model may fail to accurately characterize these properties. However, as depicted in Fig. 3, linear EH can be used to calculate the harvested power for a certain operating regime.*

Since the tag is passive, its harvested power, p_h, should exceed a minimum threshold to maintain its operation, i.e., $p_h \geq p_b$, which is about $-20\,\text{dBm}$ for commercial passive tags [15], [54]. In particular, $p_k^{\text{in}} \geq p_b'$, where $p_b' \triangleq \Phi^{-1}(p_b)$ for nonlinear EH model and $p_b' = p_b/\eta_b$ for linear EH case. Hence, our proposed algorithms can handle both linear EH and nonlinear EH models.

C. TRANSMISSION MODEL

The BS communicates with the user by sending a signal $s(n)$, where $\mathbb{E}\{|s(n)|^2\} = 1$, and the transmit power of p_t. The BS signal at the n-th time slot is thus given as

$$x(n) = \sqrt{p_t}s(n). \tag{5}$$

The tags utilize $x(n)$ to harvest energy and to communicate their data, $c_i(n)$, where $\mathbb{E}\{|c_i(n)|^2\} = 1$ for $i \in \mathcal{K}$. The user receives the BS signal and the backscattered signals from the tags. It is assumed that the propagation delay differences

for all signals are negligible [16], [29]. Therefore, the user-received signal is given as

$$y(n) = \mathbf{f}^{\mathrm{H}}\mathbf{w}x(n) + \sum\nolimits_{i\in\mathcal{K}} \sqrt{\alpha_i}\mathbf{g}_i^{\mathrm{H}}\mathbf{w}x(n)c_i(n) + z(n), \tag{6}$$

where the first and second components in $y(n)$ represent the direct-link and the backscatter-link signals. In (6), $\mathbf{g}_i$ for $i \in \mathcal{K}$ denotes the backscatter channel between the BS, the user, and T_i. Moreover, $\mathbf{w} \in \mathbb{C}^{M\times 1}$ is the beamforming vector at the BS, which satisfies $\|\mathbf{w}\|^2 = 1$. Furthermore, $z(n)$ denotes the additive white Gaussian noise (AWGN) at the user having 0 mean and σ^2 variance, i.e., $z(n) \sim \mathcal{CN}(0, \sigma^2)$. In this paper, we assume that the primary signal $s(n)$ and the tag signals $c_i(n), i \in \mathcal{K}$ are standard circular symmetric complex Gaussian distributed $\mathcal{CN}(0, 1)$ [3].

To implement power domain NOMA, without loss of generality, we assume that tag indices are sorted in decreasing channel gains, i.e., $\alpha_1|\mathbf{g}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_k|\mathbf{g}_k^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_K|\mathbf{g}_K^{\mathrm{H}}\mathbf{w}|^2$. The user performs SIC decoding to recover the tag's data. Specifically, the user initially decodes its own signal while treating tag signals as interference[1]. The user then subtracts the decoded signal, $s(n)$, from the received signal $y(n)$ and decodes each tag signal utilizing SIC decoding. To this end, the post-processed signal following the SIC decoding of T_k's data at the user is given as

$$\begin{aligned} y_k(n) = &\underbrace{\sqrt{\alpha_k p_t}\mathbf{g}_k^{\mathrm{H}}\mathbf{w}s(n)c_k(n)}_{\text{Desired signal}} \\ &+ \underbrace{\sqrt{p_t}\tilde{\mathbf{f}}^{\mathrm{H}}\mathbf{w}s(n)}_{\text{Residual user interference due to imperfect SIC}} \\ &+ \underbrace{\sqrt{p_t}\sum\nolimits_{i\in\mathcal{K}_k}\sqrt{\alpha_i}\tilde{\mathbf{g}}_i^{\mathrm{H}}\mathbf{w}s(n)c_i(n)}_{\text{Residual tag interference due to imperfect SIC}} \\ &+ \underbrace{\sqrt{p_t}\sum\nolimits_{i\in\mathcal{K}'_k}\sqrt{\alpha_i}\mathbf{g}_i^{\mathrm{H}}\mathbf{w}s(n)c_i(n)}_{\text{Tag interference after SIC}} + \underbrace{z(n)}_{\text{AWGN}}, \end{aligned} \tag{7}$$

where $\mathcal{K}_k = \{1, \ldots, k-1\}$ and $\mathcal{K}'_k = \{k+1, \ldots, K\}$. Moreover, CSI errors result in imperfect SIC or residual error propagation from the user and tags in (7).

D. ACHIEVABLE RATE

Herein, we derive the rates of the user and the tags. From the received signal (6), the rate of the user is computed as[2]

$$\mathcal{R}_0 = \log_2(1 + \gamma_0), \tag{8}$$

where

$$\gamma_0 = \frac{p_t|\mathbf{f}^{\mathrm{H}}\mathbf{w}|^2}{p_t\sum_{i\in\mathcal{K}}\alpha_i|\mathbf{g}_i^{\mathrm{H}}\mathbf{w}|^2 + \sigma^2}. \tag{9}$$

Next, the user performs SIC to decode the tag signals. Therefore, based on the post-processed signal (7), the rate of T_k can be expressed as

$$\mathcal{R}_k = \mathbb{E}_s\left\{\log_2(1 + \gamma_k^b)\right\}, \tag{10}$$

[1]The direct-link signal is much stronger than the backscatter signal [3].

[2]We approximate $s(n)c_i(n), i \in \mathcal{K}$ as complex Gaussian, which yields a lower bound of the rate of the user [3], [56].

where γ_k^b is the signal-to-interference-plus-noise ratio (SINR) of T_k at the user, which is given in (11). By taking the average over $s(n)$ in (10), the rate of T_k is computed as[3]

$$\mathcal{R}_k = \log_2(e)\left(-e^{\frac{1}{a_k+b_k}}\mathrm{E}_i\left(\frac{-1}{a_k+b_k}\right) + e^{\frac{1}{b_k}}\mathrm{E}_i\left(\frac{-1}{b_k}\right)\right), \tag{12}$$

where

$$a_k \triangleq \frac{\alpha_k p_t}{\sigma^2}|\mathbf{g}_k^{\mathrm{H}}\mathbf{w}|^2, \tag{13a}$$

$$b_k \triangleq \frac{p_t}{\sigma^2}\left(|\tilde{\mathbf{f}}^{\mathrm{H}}\mathbf{w}|^2 + \sum_{i\in\mathcal{K}_k}\alpha_i|\tilde{\mathbf{g}}_i^{\mathrm{H}}\mathbf{w}|^2 + \sum_{i\in\mathcal{K}'_k}\alpha_i|\mathbf{g}_i^{\mathrm{H}}\mathbf{w}|^2\right), \tag{13b}$$

Additionally, $\mathrm{E}_i(x) = \int_{-\infty}^{x} u^{-1}e^u du$ is the exponential integral function. Note that, $-e^{\frac{1}{x}}\mathrm{E}_i(-1/x)$ is monotonically increasing and concave function of x [3].

Furthermore, according to Section II-B, the tags must exceed the power threshold. Thus, the linear EH model (3) implies that for T_k to function:

$$(1 - \alpha_k)p_k^a \geq p_b', \tag{14}$$

where $p_k^a = p_t|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}|^2$ is the received power at T_k antenna.

III. OPTIMIZATION PROBLEM FORMULATION

Herein, we optimize the performance of the considered network (Fig. 2). The optimization variables are the BS beamforming vector (i.e., $\mathbf{w}$), tags' reflection coefficients (i.e., $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_K]$), and the BS transmit power (i.e., p_t). We utilize two optimization criteria: (i) maximize the weighted network sum rate and (ii) minimize the BS transmit power. We must guarantee that all nodes, including the user and the tags, achieve a minimum rate and that the tags meet the minimum operational energy requirements. To this end, we formulate the following resource allocation schemes:

1) We optimize $\mathbf{w}$ and $\boldsymbol{\alpha}$ to maximize the weighted sum rate of the system, i.e., rates of the primary user and tags, under minimum rate requirements for all devices and EH requirements at the tags,
2) We minimize p_t by optimizing $\mathbf{w}$ and $\boldsymbol{\alpha}$ while satisfying minimum rate requirements for all devices and EH requirements at the tags.

These two problems address two critical concerns to enabling ambient IoT. The first improves the user and tag rates. The latter increases the number of connected tags for realizing large-scale ambient IoT networks.

The optimization is carried out at the BS, which establishes communication with the tags through a control link. Tag reflection coefficients, $\boldsymbol{\alpha}$, affect not only the tag rate (10) and the harvested power at the tags (14) but also the rate of the primary user (8). Therefore, the optimal design of $\boldsymbol{\alpha}$ will significantly improve the performance of the two networks. Nevertheless, to quantify the gains of both fixed and variable $\boldsymbol{\alpha}$, we develop resource allocation for the network considering two distinct scenarios:

[3]For $s(n) \sim \mathcal{CN}(0, 1)$, $|s(n)|^2$ is exponentially distributed.

$$\gamma_k^b = \frac{\alpha_k p_t |\mathbf{g}_k^{\mathrm{H}}\mathbf{w}|^2 |s(n)|^2}{p_t\left(|\tilde{\mathbf{f}}^{\mathrm{H}}\mathbf{w}|^2 + \sum_{i\in\mathcal{K}_k}\alpha_i|\tilde{\mathbf{g}}_i^{\mathrm{H}}\mathbf{w}|^2 + \sum_{i\in\mathcal{K}_k'}\alpha_i|\mathbf{g}_i^{\mathrm{H}}\mathbf{w}|^2\right)|s(n)|^2 + \sigma^2} \tag{11}$$

1) Passive tags with fixed reflection coefficients,
2) Reconfigurable tags with variable reflection coefficients.

For case 1, the BS does not require transmitting $\boldsymbol{\alpha}$ value, but other forms of communication are necessary. This is in accordance with RFID standards, which demonstrate how such communication can be achieved. In this setup, the reader communicates with passive tags by modulating an RF carrier, enabling operations like reading, writing, locking, or killing the tags by transmitting appropriate commands [57].

On the other hand, for case 2, where tags need to dynamically modify their reflection coefficients, a different approach is followed. Within each coherence interval of the channel, the BS identifies the optimal values of α and then sends them to the tags via the control link. Since reconfigurable tags have a more advanced tag architecture than passive tags, they can receive and process data from the BS [15]–[17].

In the following two subsections, we lay out the fundamentals necessary for handling optimization problems involving both passive and reconfigurable tags.

A. WEIGHTED SUM RATE MAXIMIZATION

Many resource allocation problems directly or indirectly rely on the weighted sum rate criterion [58]. It is thus an important performance objective for a general wireless network with arbitrary interfering links to model different trade-offs between sum rate and user fairness. That is why we consider it here under the user and tag rate requirements, the tag's power threshold, and the BS power constraint. This optimization thus enhances the whole network while maintaining sufficient user and tag rates.

The weighted sum rate maximization problem can be formulated as

$$\begin{aligned}
\mathbf{P1}: \underset{\mathbf{w},\boldsymbol{\alpha}}{\text{maximize}} \quad & \sum\nolimits_{k\in\mathcal{K}_0} u_k\mathcal{R}_k, && \text{(15a)}\\
\text{subject to} \quad & \mathcal{R}_k \geq \mathcal{R}_k^{\text{th}}, \quad \text{for} \quad k\in\mathcal{K}_0, && \text{(15b)}\\
& (1-\alpha_k)|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p_b', \text{ for } k\in\mathcal{K}, && \text{(15c)}\\
& \|\mathbf{w}\|^2 = 1, && \text{(15d)}\\
& \alpha_1|\mathbf{g}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_K|\mathbf{g}_K^{\mathrm{H}}\mathbf{w}|^2, && \text{(15e)}\\
& 0<\alpha_k<1, \text{ for } k\in\mathcal{K}, && \text{(15f)}
\end{aligned}$$

where $\mathcal{K}_0 = \{0,\ldots,K\}$, $u_k \in [0,1]$ for $k\in\mathcal{K}_0$ is the weight factor and $\sum_{k\in\mathcal{K}_0} u_k = 1$. Note that the constraint in (15b) guarantees the required rate for the user and the tags, in which $R_k^{\text{th}}, k\in\mathcal{K}_0$ denotes the minimal rates of user and the tags, and R_k, $k\in\mathcal{K}_0$ is given in (8) and (10), respectively. Besides, (15c) is the tag's power threshold, and (15d) normalizes the beamforming vector.

B. TRANSMIT POWER MINIMIZATION

While weighted sum rate maximization holds critical importance, it tends to overlook green energy considerations. Among the essential metrics, power consumption has garnered significant attention. High power usage at the BS can restrict the connectivity of devices in cellular networks. As the number of connected devices grows, particularly for establishing massive machine-type communication (mMTC), the reduction of power consumption while ensuring applications meet the minimum rate requirement becomes paramount [59].

Thus, as an alternative to weighted sum rate maximization, we aim to minimize the BS transmit power. This is achieved by optimizing the transmit beamforming vector $\mathbf{w}$ and $\boldsymbol{\alpha}$, while also ensuring that the user and tag meet the minimum rate requirements, and the tag EH constraints are satisfied.

The transmit power minimization problem can be formulated as

$$\begin{aligned}
\mathbf{P2}: \underset{\mathbf{w},p_t,\boldsymbol{\alpha}}{\text{minimize}} \quad & p_t, && \text{(16a)}\\
\text{subject to} \quad & \mathcal{R}_k \geq \mathcal{R}_k^{\text{th}}, \quad \text{for} \quad k\in\mathcal{K}_0, && \text{(16b)}\\
& (1-\alpha_k)|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p_b', \text{ for } k\in\mathcal{K}, && \text{(16c)}\\
& \|\mathbf{w}\|^2 = 1, && \text{(16d)}\\
& \alpha_1|\mathbf{g}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_K|\mathbf{g}_K^{\mathrm{H}}\mathbf{w}|^2, && \text{(16e)}\\
& 0<\alpha_k<1, \text{ for } k\in\mathcal{K}. && \text{(16f)}
\end{aligned}$$

Remark 2. *In order to solve the weighted sum rate maximization and transmit power minimization problems, we replace the rate of each tag, given in* (10)*, with its lower bound for tractability. To derive the lower bound, using* (10)

$$\begin{aligned}
\mathcal{R}_k &= \mathbb{E}_x\left\{\log_2\left(\frac{1+(a_k+b_k)x(n)}{b_k x(n)+1}\right)\right\},\\
&\geq \underbrace{\mathbb{E}_x\left\{\log_2\left(1+(a_k+b_k)x(n)\right)\right\}}_{\textit{Lower bound}}\\
&\quad - \underbrace{\mathbb{E}_x\left\{\log_2\left(1+b_k x(n)\right)\right\}}_{\textit{Upper bound}},\\
&\overset{(a)}{=} \log_2\left(1+\frac{(a_k+b_k)}{2}\right) - \log_2\left(1+b_k\right)\\
&= \log_2\left(1+\frac{(a_k-b_k)}{2(1+b_k)}\right) \triangleq \mathcal{R}_k^{\text{lb}},
\end{aligned} \tag{17}$$

where a_k and b_k are given in (13)*, and $x(n) = |s(n)|^2$. In* (17)*, (a) is obtained by using [60]*

$$\mathcal{F}_{\text{lb}} \leq \mathbb{E}_x\{\log_2(1+f(x))\} \leq \mathcal{F}_{\text{ub}}. \tag{18}$$

where $\mathcal{F}_{\text{ub}} = \log_2(1+\mathbb{E}_x\{f(x)\})$, and $\mathcal{F}_{\text{lb}} = \log_2(1+[\mathbb{E}_x\{1/f(x)\}]^{-1})$, in which $\mathbb{E}_x\{1/f(x)\} = 1/\mathbb{E}_x\{f(x)\} + \sigma^2_{f(x)}/[\mathbb{E}_x\{f(x)\}]^3$. Note that since the BS performs the optimization with the estimated CSI of the channels, all of

the objective functions and constraints in $\mathbf{P1}$ *and* $\mathbf{P2}$ *are replaced with their respective channel estimations* (2).

We note that the weighted sum rate maximization and transmit power minimization problems, $\mathbf{P1}$ and $\mathbf{P2}$, are non-convex since the objective function of $\mathbf{P1}$ and the constraints of $\mathbf{P1}$ and $\mathbf{P2}$ are not convex functions in $\mathbf{w}$, p_t, and $\boldsymbol{\alpha}$. Non-convex problems are in general hard to solve optimally. Besides, the joint optimization of $\mathbf{w}$ and $\boldsymbol{\alpha}$ is complex. Therefore, we decouple the optimization variables and resort to AO, FP, and SDR techniques to solve the problems.

The basic insight of AO is to split the optimization variables into distinct non-overlapping blocks of one or more variables. By doing so, we can efficiently optimize the blocks one at a time in an iterative manner. During each iteration, we focus on optimizing a single block while keeping the remaining blocks fixed, proceeding to the next block in a sequential manner until convergence is reached. This effective optimization strategy is also known as block coordinate descent [44].

In our problem, there is a natural choice of blocks, namely $\mathbf{w}, \boldsymbol{\alpha}$ and p_t. We can thus define several sub-problems to optimize these blocks individually. Clearly, this division can simplify the overall optimization and may enhance the convergence process. AO allows us to concentrate on individual blocks, each with its own optimization algorithm tailored to its specific characteristics. For that, we will be using the following optimization techniques.

SDR is a highly efficient method for approximating and solving Quadratically Constrained Quadratic Programs (QCQP) with lower computational complexity. By converting the original QCQP into a Semi-Definite Programming (SDP) with a non-convex rank one constraint, SDR can provide an exact optimal solution to the original problem by relaxing the rank one constraint [43]. On the other hand, FP is a powerful approach to address optimization problems with objective functions that have a ratio of two nonlinear functions. By separating the numerator and denominator of each ratio term and transforming a concave-convex multiple ratios FP problem into a series of convex optimization problems, FP is an effective way to solve such problems [40]–[42].

By effectively combining these three approaches, we next propose and develop solutions to $\mathbf{P1}$ and $\mathbf{P2}$ for both passive tags and reconfigurable tags.

IV. PASSIVE TAGS

In this section, we only consider passive tags with limited processing capabilities. Hence, their reflection coefficients are fixed, i.e., $\alpha_k = \alpha$ for $k \in \mathcal{K}$.

A. WEIGHTED SUM RATE MAXIMIZATION

The objective and the constraints of the weighted sum rate maximization problem, $\mathbf{P1}$, are non-convex, and hence popular convex algorithms do not work. Thus, we utilize the FP paradigm to address the non-convex objective function, transforming the resultant problem into a convex form. To this end, we first convert the rate requirement constraints in (15b) into the SINR constraints, as $\mathcal{R}_k$ is a non-decreasing function of its argument. The weighted sum rate maximization problem can then be written as

$$\begin{aligned}
\mathbf{P}_{\mathrm{w}} : \underset{\mathbf{w}}{\text{maximize}} \ & \sum\nolimits_{k\in\mathcal{K}_0} u_k \hat{\mathcal{R}}_k, && \text{(19a)}\\
\text{subject to} \ & \hat{\gamma}_k \geq \gamma_k^{\mathrm{th}}, \quad \text{for} \quad k \in \mathcal{K}_0, && \text{(19b)}\\
& (1-\alpha)|\hat{\mathbf{h}}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p_b', \ \text{for } k \in \mathcal{K}, && \text{(19c)}\\
& \|\mathbf{w}\|^2 = 1, && \text{(19d)}\\
& |\hat{\mathbf{g}}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq |\hat{\mathbf{g}}_K^{\mathrm{H}}\mathbf{w}|^2, && \text{(19e)}
\end{aligned}$$

where $\gamma_k^{\mathrm{th}} = 2^{R_k^{\mathrm{th}}} - 1$. Here, $\hat{\mathcal{R}}_k$ and $\hat{\gamma}_k$ are obtained by replacing the respective channels $\mathbf{f}$ and $\mathbf{g}_k, k \in \mathcal{K}$ with (2). Moreover, $\hat{\gamma}_k = (\hat{a}_k - \hat{b}_k)/(2 + 2\hat{b}_k)$ for $k \in \mathcal{K}$ is also by substituting the respective actual channels in (17) with their channel estimates.

To solve (19), we introduce an auxiliary variable $\beta_k, k \in \mathcal{K}_0$ to replace the SINR terms in the objective function (19a) such that $\beta_k \leq \gamma_k$ and then reformulate $\mathbf{P}_{\mathrm{w}}$ (19) as

$$\begin{aligned}
\mathbf{P}_{\mathrm{w1}} : \underset{\mathbf{w},\boldsymbol{\beta}}{\text{maximize}} \ & \sum\nolimits_{k\in\mathcal{K}_0} u_k \log_2(1+\beta_k), && \text{(20a)}\\
\text{subject to} \ & \gamma_k^{\mathrm{th}} \leq \beta_k \leq \frac{\hat{A}_k(\mathbf{w})}{\hat{B}_k(\mathbf{w})}, \quad \text{for} \quad k \in \mathcal{K}_0, && \text{(20b)}\\
& (1-\alpha)|\hat{\mathbf{h}}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p_b', \ \text{for } k \in \mathcal{K}, && \text{(20c)}\\
& \|\mathbf{w}\|^2 = 1, && \text{(20d)}\\
& |\hat{\mathbf{g}}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq |\hat{\mathbf{g}}_K^{\mathrm{H}}\mathbf{w}|^2, && \text{(20e)}
\end{aligned}$$

where $\boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_K]^{\mathrm{T}}$. Besides, $\hat{A}_k(x)$ and $\hat{B}_k(x)$ are the numerators and denominators of the corresponding SINR terms $\hat{\gamma}_k, k \in \mathcal{K}_0$ as functions of the variable x (i.e., in here the variable is $\mathbf{w}$). The reformulated optimization problem $\mathbf{P}_{\mathrm{w1}}$ can be considered as a two-part optimization problem, i.e., (i) an outer optimization over $\mathbf{w}$ with fixed $\boldsymbol{\beta}$ and (ii) an inner optimization over $\boldsymbol{\beta}$ with fixed $\mathbf{w}$.

The inner optimization problem is thus given as

$$\begin{aligned}
\mathbf{P}_{\mathrm{w2}} : \underset{\boldsymbol{\beta}}{\text{maximize}} \ & \sum\nolimits_{k\in\mathcal{K}_0} u_k \log_2(1+\beta_k), && \text{(21a)}\\
\text{subject to} \ & \gamma_k^{\mathrm{th}} \leq \beta_k \leq \frac{\hat{A}_k(\mathbf{w})}{\hat{B}_k(\mathbf{w})}, \quad \text{for} \quad k \in \mathcal{K}_0, && \text{(21b)}
\end{aligned}$$

The inner optimization problem in (21) is a convex optimization in $\boldsymbol{\beta}$ and holds strong duality [41]. Therefore, the solution to $\mathbf{P}_{\mathrm{w2}}$ is that β_k satisfies (21b) with equality. To deal with the logarithm in the objective function of $\mathbf{P}_{\mathrm{w2}}$, we apply the Lagrangian dual transform [41], and the corresponding Lagrangian function can be written as

$$L(\boldsymbol{\beta}, \boldsymbol{\lambda}) = \sum_{k\in\mathcal{K}_0} u_k \log_2(1+\beta_k) - \sum_{k\in\mathcal{K}_0} \lambda_k \left(\beta_k - \frac{\hat{A}_k(\mathbf{w})}{\hat{B}_k(\mathbf{w})}\right), \quad (22)$$

where $\boldsymbol{\lambda} = [\lambda_0, \lambda_1, \ldots, \lambda_K]^{\mathrm{T}}$ is the dual variable vector introduced for each inequality constraint in (21b). From the

strong duality, $\mathbf{P}_{\mathrm{w2}}$ can be equivalently reformulated to a dual problem as

$$\mathbf{P}_{\mathrm{w3}}: \underset{\boldsymbol{\lambda}\succeq 0}{\text{minimize}} \quad \underset{\boldsymbol{\beta}}{\text{maximize}} \quad L(\boldsymbol{\beta},\boldsymbol{\lambda}). \tag{23}$$

By evaluating the first-order condition $\partial L(\boldsymbol{\beta},\boldsymbol{\lambda})/\partial\beta_k, k \in \mathcal{K}_0$ and using the trivial solution to $\mathbf{P}_{\mathrm{w2}}$, the optimal solution of $\lambda_k, k \in \mathcal{K}_0$ can be obtained as

$$\lambda_k^o = \frac{u_k \hat{B}_k(\mathbf{w})}{\hat{A}_k(\mathbf{w}) + \hat{B}_k(\mathbf{w})}. \tag{24}$$

Besides, $\lambda_k \geq 0$ is automatically satisfied in this case. Using (22) and (24), $\mathbf{P}_{\mathrm{w2}}$ can be reformulated as

$$\mathbf{P}_{\mathrm{w4}}: \underset{\boldsymbol{\beta}}{\text{maximize}} \quad L(\boldsymbol{\beta},\boldsymbol{\lambda}^o). \tag{25}$$

Furthermore, it can be proved that when coupled with the outer optimization over $\mathbf{w}$ and after several mathematical interpretations, the solution to $\mathbf{P}_{\mathrm{w4}}$ also satisfies $\mathbf{P}_{\mathrm{w}}$ (20) [41], [61]. Then, the beamforming vector, $\mathbf{w}$, is obtained by solving feasibility problem over $\mathbf{w}$ for fixed $\boldsymbol{\beta}$ (20). We give our solution to the weighted sum rate problem $\mathbf{P}_{\mathrm{w}}$, regarding the above formulation, in Algorithm 1.

Algorithm 1 : Transmit beamforming.

Initialization: Initialize $\mathbf{w}$ to a feasible value.
Repeat
Step 1: Update $\boldsymbol{\lambda}$ by (24).
Step 2: Update $\boldsymbol{\beta}$ by solving $\mathbf{P}_{\mathrm{w4}}$ in (25).
Step 3: Update $\mathbf{w}$ by solving feasibility problem over $\mathbf{w}$ for fixed $\boldsymbol{\beta}$, $\mathbf{P}_{\mathrm{w}}$.
Until the value of the objective function converges.
Output: The optimal beamforming vector $\mathbf{w}^o$.

Remark 3. *Algorithm 1 describes the proposed optimization methods for solving* $\mathbf{w}$*, in which we utilize FP to transform* $\mathbf{P}_{\mathrm{w}}$ *into a convex optimization problem. We begin by quantifying the SINR of the user and* T_k*,* $k \in \mathcal{K}$*, as we initialize* $\mathbf{w}$ *to a feasible value, and then in each iteration, a better solution for* $\mathbf{w}$ *is updated by computing* $\boldsymbol{\beta}$ *and* $\boldsymbol{\lambda}$*. The procedure continues until the objective stops increasing. The normalized objective function increment is less than* $\epsilon = 10^{-3}$*.*

B. TRANSMIT POWER MINIMIZATION

To solve the transmit power minimization problem $\mathbf{P2}$, similar to $\mathbf{P1}$, we first convert the rate requirement constraints in (16b) into the SINR constraints and reformulate $\mathbf{P2}$ as

$$\begin{aligned}
\mathbf{P}_{\mathrm{p}}: \underset{\mathbf{w},p_t}{\text{minimize}} \;& p_t, && (26a)\\
\text{subject to} \;& \hat{\gamma}_k \geq \gamma_k^{\mathrm{th}}, \quad \text{for} \quad k \in \mathcal{K}_0, && (26b)\\
& (1-\alpha)|\hat{\mathbf{h}}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p_b', \text{ for } k \in \mathcal{K}, && (26c)\\
& \|\mathbf{w}\|^2 = 1, && (26d)\\
& |\hat{\mathbf{g}}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq |\hat{\mathbf{g}}_K^{\mathrm{H}}\mathbf{w}|^2. && (26e)
\end{aligned}$$

Problem $\mathbf{P}_{\mathrm{p}}$ is a non-convex optimization since the constraints are non-convex with respect to $\mathbf{w}$. Further, this quadratic problem is NP-hard [43]. We thus utilize the SDR technique to solve $\mathbf{P}_{\mathrm{p}}$.

The SDR of problem $\mathbf{P}$ is obtained by relaxing some of its constraints and extending the objective function's domain [62]. Therefore, the optimal SDR value represents an upper bound on the optimal value of $\mathbf{P}$ [62]. Here, we relax the non-convex rank one constraint, i.e., the only challenging constraint to deal with SDP, and employ SDP to solve the relaxed problem.

We first define $\mathbf{v} \triangleq \sqrt{p_t}\mathbf{w}$, $\mathbf{W} \triangleq \mathbf{v}\mathbf{v}^{\mathrm{H}}$, and $\mathbf{A} \triangleq \mathbf{a}\mathbf{a}^{\mathrm{H}}$, where $\mathbf{a} \in \{\mathbf{f}, \mathbf{g}_k\}, k \in \mathcal{K}$, and rewrite the SINR (9) as

$$\gamma_0 = \frac{\mathrm{Tr}(\mathbf{F}\mathbf{W})}{\alpha \sum_{i\in\mathcal{K}} \mathrm{Tr}(\mathbf{G}_i\mathbf{W}) + \sigma^2}, \tag{27}$$

and (11) as (28), shown in the top of the next page. Next, we replace the respective true channels $\mathbf{f}$ and $\mathbf{g}_k, k \in \mathcal{K}$ in (27) and (28) with (2). Thereby, the problem, $\mathbf{P}_{\mathrm{p}}$, is reformulated into the following equivalent problem:

$$\begin{aligned}
\mathbf{P}_{\mathrm{p1}}: \underset{\mathbf{W}}{\text{minimize}} \;& \mathrm{Tr}(\mathbf{W}), && (29a)\\
\text{subject to} \;& \hat{\gamma}_k \geq \gamma_k^{\mathrm{th}}, \quad \text{for} \quad k \in \mathcal{K}_0, && (29b)\\
& (1-\alpha)\,\mathrm{Tr}(\hat{\mathbf{H}}_k\mathbf{W}) \geq p_b', \text{ for } k \in \mathcal{K}, && (29c)\\
& \mathrm{Tr}(\mathbf{W}) \leq p_{\max}, && (29d)\\
& \mathrm{Rank}(\mathbf{W}) = 1, && (29e)\\
& \mathrm{Tr}(\hat{\mathbf{G}}_1\mathbf{W}) \geq \ldots \geq \mathrm{Tr}(\hat{\mathbf{G}}_K\mathbf{W}), && (29f)
\end{aligned}$$

where $p_{\max}$ denotes the maximum allowable transmit power at the BS. Because the rank-one constraint in (29e) is highly non-convex, relaxing it allows $\mathbf{P}_{\mathrm{p1}}$ to be reformulated as the following SDR problem [3], [43]:

$$\begin{aligned}
\mathbf{P}_{\mathrm{p2}}: \underset{\mathbf{W}}{\text{minimize}} \;& \mathrm{Tr}(\mathbf{W}), && (30a)\\
\text{subject to} \;& \hat{\gamma}_k \geq \gamma_k^{\mathrm{th}}, \quad \text{for} \quad k \in \mathcal{K}_0, && (30b)\\
& (1-\alpha)\,\mathrm{Tr}(\hat{\mathbf{H}}_k\mathbf{W}) \geq p_b', \text{ for } k \in \mathcal{K}, && (30c)\\
& \mathrm{Tr}(\mathbf{W}) \leq p_{\max}, && (30d)\\
& \mathrm{Tr}(\hat{\mathbf{G}}_1\mathbf{W}) \geq \ldots \geq \mathrm{Tr}(\hat{\mathbf{G}}_K\mathbf{W}). && (30e)
\end{aligned}$$

This problem, $\mathbf{P}_{\mathrm{p2}}$, is the SDR of $\mathbf{P}_{\mathrm{p1}}$, which is an SDP problem. Moreover, $\mathbf{P}_{\mathrm{p2}}$ is a convex optimization problem that can be handled using SDP in CVX Matlab [43], [63]. If the solution to this relaxed-SDR problem, $\mathbf{P}_{\mathrm{p2}}$, is of rank one, i.e., $\mathbf{W}^o = \mathbf{v}^o(\mathbf{v}^o)^{\mathrm{H}}$, the optimal solution to $\mathbf{P}_{\mathrm{p1}}$ is $p_t^o = \mathrm{Tr}(\mathbf{S})$ and $\mathbf{w}^o = \mathbf{v}^o/\sqrt{p_t^o}$, where $\mathbf{S}$ is obtained by evaluating the singular value decomposition (SVD) of $\mathbf{W}^o$ as $\mathbf{W}^o = \mathbf{U}\mathbf{S}\mathbf{U}^{\mathrm{H}}$. Otherwise, to account for the relaxed rank-one constraint, we use Gaussian randomization-based approach to find an approximate solution to $\mathbf{P}_{\mathrm{p1}}$ [64]. Algorithm 2 summarizes the steps required to solve $\mathbf{P}_{\mathrm{p1}}$.

Remark 4. *The proposed transmit power minimization problem is first transformed into a non-convex rank-one SDR problem. By relaxing this rank-one constraint, we solve the*

$$\gamma_k = \frac{\alpha \operatorname{Tr}(\mathbf{G}_k\mathbf{W}) - \left(\operatorname{Tr}(\tilde{\mathbf{F}}\mathbf{W}) + \alpha \sum_{i\in\mathcal{K}_k} \operatorname{Tr}(\tilde{\mathbf{G}}_i\mathbf{W}) + \alpha \sum_{i\in\mathcal{K}'_k} \operatorname{Tr}(\mathbf{G}_i\mathbf{W})\right)}{2\left(\operatorname{Tr}(\tilde{\mathbf{F}}\mathbf{W}) + \alpha \sum_{i\in\mathcal{K}_k} \operatorname{Tr}(\tilde{\mathbf{G}}_i\mathbf{W}) + \alpha \sum_{i\in\mathcal{K}'_k} \operatorname{Tr}(\mathbf{G}_i\mathbf{W}) + \sigma^2\right)} \tag{28}$$

Algorithm 2 : Power minimization.

Initialization: Begin - CVX.
  Step 1: Solve the convex problem $\mathbf{P}_{\mathrm{p2}}$ in (30).
  Step 2: SVD $\mathbf{W}^o$ as $\mathbf{W}^o = \mathbf{U}\mathbf{S}\mathbf{U}^{\mathrm{H}}$, where
    $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_M]$.
  End - CVX.
  if $\operatorname{Rank}(\mathbf{W}^o) = 1$, **then**
    return: $p_t^o = \operatorname{Tr}(\mathbf{S})$ and $\mathbf{w}^o = \mathbf{u}_1$.
  else
    for $d = 1, \ldots, D$ **do**
      1: Generate random $\mathbf{v}_d = \mathbf{U}\mathbf{S}^{1/2}\mathbf{e}_d$, where $\mathbf{e}_d = [\mathrm{e}^{j\phi_1}, \mathrm{e}^{j\phi_2}, \ldots, \mathrm{e}^{j\phi_M}]^{\mathrm{H}}$ and $\phi_i \sim \mathcal{U}(0, 2\pi)$.
      2: Check if $\mathbf{P}_{\mathrm{p2}}$ is feasible with $\mathbf{v}_d$.
    end for
    return: $p_t^o = \|\mathbf{v}^o\|^2$ and $\mathbf{w}^o = \mathbf{v}^o/\sqrt{p_t^o}$, where $\mathbf{v}^o = \arg\min_{d=1,\ldots,D} \mathbf{v}_d$.
  end if

Output: Optimal transmit power p_t^o and precoder $\mathbf{w}^o$.

relaxed-SDR problem with SDP and then impose the relaxed rank-one constraint with the Gaussian randomization technique. We initially set p_t and $\mathbf{w}$ to feasible values, and then update better solutions for p_t and $\mathbf{w}$ using the steps in Algorithm 2. The procedure is repeated until there is no more progress, at which point it is subjected to a termination criterion such that the increment of the normalized objective function is smaller than $\epsilon = 10^{-3}$.

C. COMPUTATIONAL COMPLEXITY

The proposed approach for solving the weighted sum rate maximization problem is a multistage iterative algorithm, i.e., Algorithm 1, which is developed utilizing FP. In particular, the outer loop is responsible for optimizing $\mathbf{w}$, while the inner loop is responsible for updating the auxiliary variables, $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$, introduced by FP techniques. The computational complexity of Algorithm 1 lies in step 3. Additionally, CVX Matlab uses an SDPT3 solver to handle this optimization problem. Therefore, the computational complexity of Algorithm 1 is $\mathcal{O}((K+1)^3M^3)$ [40], [65]. Hence, the proposed solution for the weighted sum rate maximization has a total complexity of $\mathcal{O}(I_w(K+1)^3M^3)$, where I_w is the iteration numbers of Algorithm 1.

In $\mathbf{P}_{\mathrm{p2}}$, the SDR problem is traditional SDP, which is solved using well-known interior point techniques [66], [67]. For Algorithm 2, given the number of constraints, the interior point techniques with a matrix $\mathbf{W}$ with variables of size $\mathcal{O}(\sqrt{M}\log(1/\kappa))$ will need $\mathcal{O}(\sqrt{M}\log(1/\kappa))$ iterations, with each iteration requiring $\mathcal{O}(M^6)$ arithmetic operations in the worst case, where κ is the accuracy of the interior-point method [3], [66], [67].

V. RECONFIGURABLE TAGS

Herein, we investigate the effect of optimal tags' reflection coefficients. To this end, we maximize the weighted sum rate of the system by optimizing $\mathbf{w}$ and $\boldsymbol{\alpha}$. We omit the transmit power minimization for brevity.

A. WEIGHTED SUM RATE MAXIMIZATION

Because the optimization variables in $\mathbf{P1}$ are coupled, the joint optimization of $\mathbf{w}$ and $\boldsymbol{\alpha}$ is complex. Hence, we decouple $\mathbf{w}$ and $\boldsymbol{\alpha}$, yielding two sub-problems. To that end, $\mathbf{P1}$ reduces to the transmit beamforming optimization problem, $\mathbf{P}_{\mathrm{w}'}$, which is similar to $\mathbf{P}_{\mathrm{w}}$ with respect to fixed α_k's.

Next, $\mathbf{P1}$ becomes an optimization problem for tags' reflection coefficients for a given $\mathbf{w}$. The corresponding optimization problem is given as follows:

$$\mathbf{P}_\alpha : \underset{\boldsymbol{\alpha}}{\text{maximize}} \sum\nolimits_{k\in\mathcal{K}_0} u_k \log_2(1+\hat{\gamma}_k), \tag{31a}$$
$$\text{subject to } \hat{\gamma}_k \geq \gamma_k^{\mathrm{th}}, \quad \text{for} \quad k \in \mathcal{K}_0, \tag{31b}$$
$$(1-\alpha_k)|\hat{\mathbf{h}}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p'_b, \text{ for } k \in \mathcal{K}, \tag{31c}$$
$$\alpha_1|\mathbf{g}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_K|\mathbf{g}_K^{\mathrm{H}}\mathbf{w}|^2, \tag{31d}$$
$$0 < \alpha_k < 1, \text{ for } k \in \mathcal{K}. \tag{31e}$$

Because of the non-convexity of the corresponding objective functions and constraints, $\mathbf{P}_{\mathrm{w}'}$ and $\mathbf{P}_\alpha$ are not convex. We use the AO technique to alternately maximizes $\mathbf{P}_{\mathrm{w}'}$ and $\mathbf{P}_\alpha$ until the weighted sum rate objective converges. Because $\mathbf{P}_{\mathrm{w}'}$ is similar to $\mathbf{P}_{\mathrm{w}}$ (i.e., Algorithm 1 in Section IV-A can be used with slight modifications), we omit the details for brevity.

To solve $\mathbf{P}_\alpha$, we first introduce θ_k to replace the SINR terms in (31a) such that $\theta_k \leq \hat{\gamma}_k$, and reformulate $\mathbf{P}_\alpha$ as

$$\mathbf{P}_{\alpha 1} : \underset{\boldsymbol{\alpha},\boldsymbol{\theta}}{\text{maximize}} \sum\nolimits_{k\in\mathcal{K}_0} u_k \log_2(1+\theta_k), \tag{32a}$$
$$\text{subject to } \gamma_k^{\mathrm{th}} \leq \theta_k \leq \frac{\hat{A}_k(\boldsymbol{\alpha})}{\hat{B}_k(\boldsymbol{\alpha})}, \quad \text{for} \quad k \in \mathcal{K}_0, \tag{32b}$$
$$(1-\alpha_k)|\hat{\mathbf{h}}_k^{\mathrm{H}}\mathbf{w}|^2 p_t \geq p'_b, \text{ for } k \in \mathcal{K}, \tag{32c}$$
$$\alpha_1|\mathbf{g}_1^{\mathrm{H}}\mathbf{w}|^2 \geq \ldots \geq \alpha_K|\mathbf{g}_K^{\mathrm{H}}\mathbf{w}|^2, \tag{32d}$$
$$0 < \alpha_k < 1, \text{ for } k \in \mathcal{K}. \tag{32e}$$

where $\boldsymbol{\theta} = [\theta_0, \theta_1, \ldots, \theta_K]^{\mathrm{T}}$. Like $\mathbf{P}_{\mathrm{w}1}$, $\mathbf{P}_{\alpha 1}$ becomes a two-part optimization problem, and hence for a fixed $\boldsymbol{\alpha}$, the inner optimization problem is given as

$$\mathbf{P}_{\alpha 2} : \underset{\boldsymbol{\alpha},\boldsymbol{\theta}}{\text{maximize}} \sum\nolimits_{k\in\mathcal{K}_0} u_k \log_2(1+\theta_k), \tag{33a}$$
$$\text{subject to } \gamma_k^{\mathrm{th}} \leq \theta_k \leq \frac{\hat{A}_k(\boldsymbol{\alpha})}{\hat{B}_k(\boldsymbol{\alpha})}, \quad \text{for} \quad k \in \mathcal{K}_0. \tag{33b}$$

Next, we formulate the associated Lagrangian function by introducing a dual variable, $\boldsymbol{\mu}$, as follows:

$$L(\boldsymbol{\theta},\boldsymbol{\mu})=\sum_{k\in\mathcal{K}_0}u_k\log_2(1+\theta_k)-\sum_{k\in\mathcal{K}_0}\mu_k\left(\theta_k-\frac{\hat{A}_k(\boldsymbol{\alpha})}{\hat{B}_k(\boldsymbol{\alpha})}\right),\quad(34)$$

where $\boldsymbol{\mu}) = [\mu_0,\mu_1,\ldots,\mu_K]^{\mathrm{T}}$. From strong duality, $\mathbf{P}_{\alpha 2}$ is equivalently reformulated as

$$\mathbf{P}_{\alpha 3}: \underset{\boldsymbol{\mu}\succeq 0}{\text{minimize}} \quad \underset{\boldsymbol{\theta}}{\text{maximize}} \quad L(\boldsymbol{\theta},\boldsymbol{\mu}). \quad (35)$$

Next, from the first-order condition, $\partial L(\boldsymbol{\theta},\boldsymbol{\mu})/\partial\theta_k = 0$ for $k\in\mathcal{K}_0$, the optimal μ_k is obtained as

$$\mu_k^o=\frac{u_k\hat{B}_k(\boldsymbol{\alpha})}{\hat{A}_k(\boldsymbol{\alpha})+\hat{B}_k(\boldsymbol{\alpha})}\geq 0. \quad (36)$$

Then, $\mathbf{P}_{\alpha 2}$ is reformulated as

$$\mathbf{P}_{\alpha 4}: \underset{\boldsymbol{\theta}}{\text{maximize}} \quad L(\boldsymbol{\theta},\boldsymbol{\mu}^o). \quad (37)$$

Algorithm 3 provides the proposed approach for optimizing the tags' reflection coefficients.

Algorithm 3 : Reflection coefficient optimization.

Initialization: Initialize $\boldsymbol{\alpha}$ to a feasible value.
Repeat
Step 1: Update $\boldsymbol{\mu}$ by (36).
Step 2: Update $\boldsymbol{\theta}$ by solving $\mathbf{P}_{\alpha 4}$ in (37).
Step 3: Update $\boldsymbol{\alpha}$ by solving feasibility problem over $\boldsymbol{\alpha}$ for fixed $\boldsymbol{\theta}$ (32).
Until the value of the objective function converges.
Output: The optimal reflection coefficient $\boldsymbol{\alpha}^o$.

Remark 5. *Algorithm 1 and Algorithm 3 present the methods for solving separated sub-problems of* $\mathbf{P1}$*, i.e.,* $\mathbf{P}_{\mathrm{w}'}$ *and* $\mathbf{P}_{\alpha}$*. We begin by setting* $\mathbf{w}$ *and* $\boldsymbol{\alpha}$ *to feasible values. Then, in each iteration, algorithms update better solutions for* $\mathbf{w}$ *and* $\boldsymbol{\alpha}$ *until the objective function no longer improves. The procedure is terminated when the normalized objective function increment is less than* $\epsilon = 10^{-3}$.

B. ALGORITHM CONVERGENCE

The convergence of the AO technique is a well-established concept [44]. Specifically, as long as the individual blocks within the AO algorithm converge, the overall optimization also converges [44]. In our proposed solution for the reconfigurable tag case, we employ the FP technique to solve the individual blocks, namely $\mathbf{w}$ and $\boldsymbol{\alpha}$, in the AO algorithm. Remarkably, FP leads to a fixed-point iteration method with provable convergence [40]–[42], which assures the convergence for our proposed AO approach.

In contrast, the Gaussian randomization approach provides an approximate solution to a problem by post-processing the optimal solution of a relaxed problem, where the rank one constraint is omitted [64]. Importantly, Gaussian randomization does not introduce any additional optimization beyond solving the relaxed problem [64]. Consequently, as long as the relaxed problem converges, the use of Gaussian randomization has no impact on the overall algorithm's convergence [64].

TABLE 2: Simulation Settings

Parameter	Value	Parameter	Value
Carrier frequency	3 GHz	p_b	-20 dBm
Bandwidth	10 MHz	d_f	12 m
Noise figure	10 dB	d_{h_1}, d_{h_2}	5 m, 6 m
$u_k, k\in\mathcal{K}_0$	$1/\lvert\mathcal{K}_0\rvert$	d_{q_1}, d_{q_2}	10 m
η_b	0.6	M	32

VI. SIMULATION RESULTS

In this section, we present simulation results for assessing the performance of the NOMA-enabled BackCom system.

We have we have adopted the 3GPP urban micro (UMi) model the path-loss $\zeta_{\mathbf{a}}$ (1) with with $f_c = 3$ GHz operating frequency [68, Table B.1.2.1], where $\mathbf{a} \in \{\mathbf{f},\mathbf{h}_k,q_k\}$. Furthermore, we model the AWGN variance, σ^2, as $\sigma^2 = 10\log_{10}(N_0BN_f)$ dBm, where $N_0 = -174\,\mathrm{dBm/Hz}$, B is the bandwidth, and N_f is the noise figure. Table 2 summarizes the simulation parameters.

To evaluate the performance of the SR network and characterize the advantages of NOMA, we will consider one benchmark and the proposed system as follows:

- C_1 *SR - OMA*: A single user and a single tag network where the user first decodes their own data, then uses SIC to decode the tag signal.
- C_2 *SR - NOMA*: The proposed system setup with a single user and multiple tags. The tags are served by the proposed NOMA beamforming scheme.

We also consider two other BS beamformer benchmark for comparative evaluation purposes:

- *Random beamforming*: In multi-antenna communication systems, if the number of transmit antennas M and/or the number of users n is large and the BS has perfect CSI, a sum-rate capacity of $M\log(n)$ is achievable. However, since the BS may not have full CSI in practice, random beamforming can be implemented. Surprisingly, it achieves the same capacity asymptotically [69]. Following this concept, in our problem, $\mathbf{w}$ can be selected as a normalized complex Gaussian random vector to satisfy the NOMA constraint (15e) - or (16e). It requires no CSI.
- *Maximum ratio transmission (MRT) beamforming*: $\mathbf{w} = \mathbf{f}/\|\mathbf{f}\|$. The idea of this method is to direct the beam toward the primary user, necessitating accurate CSI. MRT beamforming thus achieves the optimal performance for the primary user without considering any interference. However, it may not be suitable for the SR network as it ignores the tags.

A. PASSIVE TAGS

1) Weighted Sum Rate Maximization

We next investigate the rate of the primary user and the tags through the proposed solution for the weighted sum

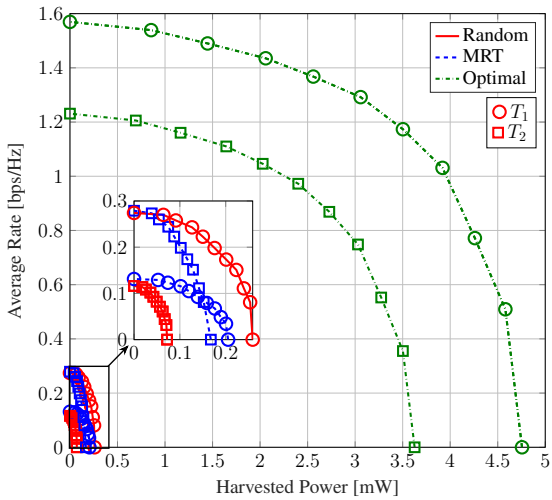

FIGURE 4: The trade-off between the harvested power and the rate at the tags for $p_t = 20\,\text{dBm}$.

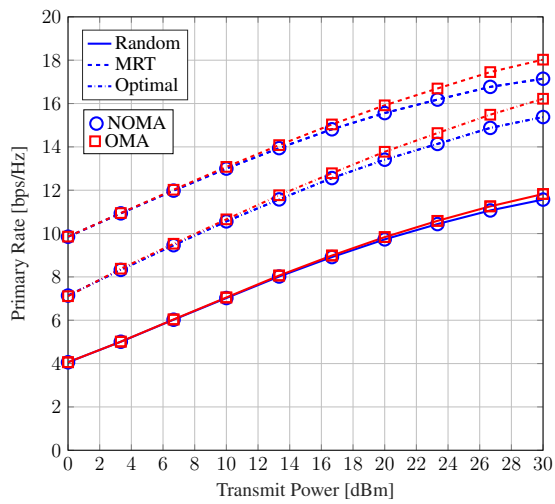

FIGURE 5: The rate of the primary user.

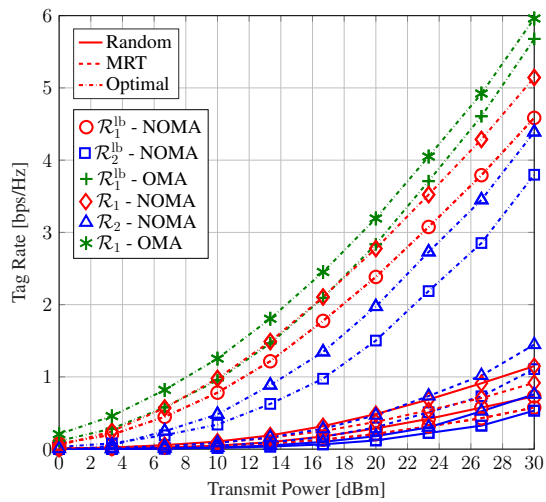

FIGURE 6: The rates of tags.

rate maximization problem (Section IV-A and Remark 3) under perfect and imperfect CSI, given in Section II-A1. We consider the rate requirements as $\mathcal{R}_0^{\text{th}} = \log_2(1 + p_t)$, $\mathcal{R}_1^{\text{th}} = \log_2(1 + p_t/50)$, and $\mathcal{R}_2^{\text{th}} = \log_2(1 + p_t/100)$. Additionally, α is set to 0.6.

Fig. 4 investigates the symbiotic benefits of the tag and the effects of the tag's reflection coefficient on the added benefits. It depicts the trade-off between harvested power and rate as a function of α for $20\,\text{dBm}$ transmit power using different transmit beamforming. When $\alpha \to 1$, the tag reflects nearly all the received power and reaches the highest rate. This, however, is not a feasible operating point for a passive tag. As $\alpha \to 0$, the tag harvests most of the received power and does not reflect any signal, resulting in an infinitesimal rate. Hence, the rate-harvested power curves corresponding to a given α provide valuable insights into the achievable trade-offs between communication rate and harvested power at the tag. Depending on specific application requirements, one can intelligently choose an appropriate α from Fig. 4. It is worth noting that as the tag operates passively, it necessitates a minimum amount of power for functioning, leading to the constraint $\alpha \in (0, 1)$. Additionally, Fig. 4 demonstrates that our proposed optimal beamforming design ensures more power delivery to both tags, enabling them to achieve higher rates. On the contrary, conventional techniques like random beamforming and MRT beamforming fail to achieve symbiotic gains for the tags.

To investigate symbiosis and the advantages of NOMA, Fig. 5 and Fig. 6 respectively show the rates of the user and the tags for BS beamforming designs and perfect CSI.

Fig. 5 depicts the primary user rate versus the transmit power. As expected, the highest rate comes from MRT beamforming, as the beam is shaped towards the user, ignoring the tags. In contrast, optimal beamforming aims to satisfy multiple goals (19). That is why it results in a lower user rate than MRT. For NOMA, at $p_t = 20\,\text{dBm}$, the user achieves $15.5\,\text{bps/Hz}$ and $13.5\,\text{bps/Hz}$ with MRT and optimal beamforming, respectively. At this power, T_1 and T_2 reach $2.9\,\text{bps/Hz}$ and $2\,\text{bps/Hz}$ with optimal beamforming (Fig. 6).

Upon observation, we notice that the user rate is slightly lower for NOMA compared to OMA due to the increased interference from tags in NOMA, see (8). Nevertheless, through optimal beamforming design, NOMA can achieve significantly higher primary user rate and tag rates than OMA. This remarkable advantage compensates for the slight degradation in user performance, allowing NOMA to support more tags and elevate the overall system performance.

Fig. 6 illustrates the lower bound rate $\mathcal{R}_k^{\text{lb}}, (k = 1, 2)$ obtained through (17), and the actual rate $\mathcal{R}_k, (k = 1, 2)$ (12) of the tags versus BS transmit power in the presence of the primary user, with different BS transmit beamforming strategies, under OMA (green curves) and NOMA (blue and red curves) scenarios. With optimal beamforming at the BS, tags achieve notably higher rates than MRT and random

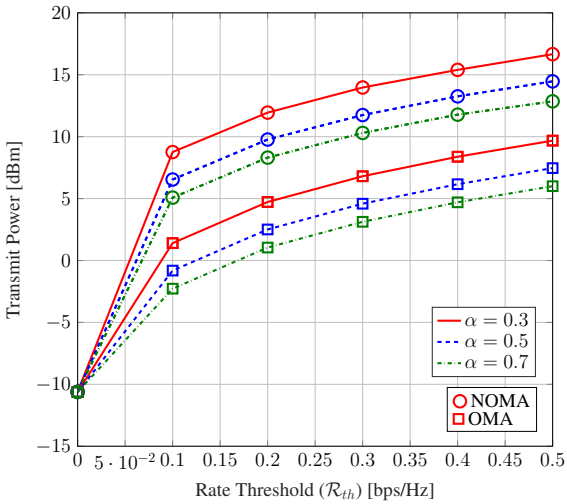

FIGURE 7: The minimum transmit power as a function of rate threshold ($\mathcal{R}_{\rm th}$).

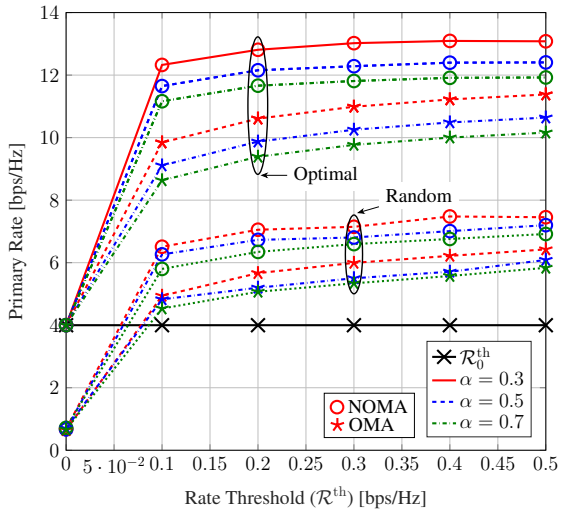

FIGURE 8: Primary user rate as a function of rate threshold ($\mathcal{R}_{\rm th}$).

beamforming in both OMA and NOMA. OMA, which serves as the benchmark, results in a superior rate for a single-tag scenario. In contrast, NOMA enables simultaneous access to the same spectrum resource by multiple tags. In particular, at $p_t = 25\,\text{dBm}$, for OMA case, the actual rate of the tag (labeled as $\mathcal{R}_1$- OMA) is $4.5\,\text{bps/Hz}$, while for the NOMA case, T_1 and T_2 achieve the rates of $3.85\,\text{bps/Hz}$ and $3.15\,\text{bps/Hz}$, respectively (labeled as $\mathcal{R}_1$- NOMA and $\mathcal{R}_2$- NOMA). Therefore, NOMA and optimal beamforming design can support more tags while achieving decent rates.

However, unlike MRT and random beamforming, our beamforming design ensures symbiosis has benefits regardless of the type of multiple access (i.e., OMA and NOMA). Notably, our beamforming design highlights the advantages of utilizing NOMA in the SR setup, achieving a significantly higher overall throughput of about $21.5\,\text{bps/Hz}$, compared to the $19.5\,\text{bps/Hz}$ achieved by OMA. This substantial throughput improvement reinforces the synergy between primary and secondary systems. In summary, our algorithm actively promotes the symbiotic relationship between primary and secondary systems, offering performance enhancements that go beyond traditional approaches like MRT and random beamforming. This results in improved overall system efficiency, making NOMA a favorable choice for SR deployment.

2) Transmit Power Minimization

Herein, we investigate the proposed transmit power minimization problem. The rate requirements for the user, T_1, and T_2 are set to $\mathcal{R}_0^{\rm th} = 4\,\text{bps/Hz}$, $\mathcal{R}_1^{\rm th} = 2\mathcal{R}_{\rm th}$, and $\mathcal{R}_2^{\rm th} = \mathcal{R}_{\rm th}$, respectively, where $\mathcal{R}_{\rm th}$ is the lowest rate requirement of a tag.

Fig. 7, Fig. 8, Fig. 9, and Fig. 10 investigate the minimum required power at the BS to support symbiosis while maintaining the user and tags rate requirements. These figures show the benefits of the proposed algorithms.

Fig. 7 illustrates the minimum transmit power as a function of the minimum tag rate threshold for various tag reflection coefficients, α, under perfect CSI. When the tags require zero rates, i.e., when the tag does not exist, the purpose is to maintain the primary user's rate requirement. As a result, in both OMA and NOMA cases, the BS transmits at $-10\,\text{dBm}$ transmit power (Fig. 7) while satisfying the primary user's minimum rate requirement (Fig. 8). Nevertheless, as the tags demand non-zero rate requirements, i.e., when the minimum tag rate threshold ($\mathcal{R}_{\rm th}$) increases, the BS must allocate more power to satisfy the rate requirements of the user and the tags simultaneously. For example, the proposed NOMA scheme with $\alpha = 0.5$ at the tags causes a transmit power of $11.7\,\text{dBm}$ to achieve the minimum rates of $0.3\,\text{bps/Hz}$ and $0.6\,\text{bps/Hz}$ for T_1 and T_2, respectively, while maintaining the primary user's rate requirement and EH requirements at the tags. This power demand increases by $2.8\,\text{dBm}$ when the minimum tag rates for T_1 and T_2 rise to $0.5\,\text{bps/Hz}$ and $1\,\text{bps/Hz}$, respectively. Compared to the benchmark scheme, OMA with only T_1, the proposed NOMA approach needs $\sim 7\,\text{dBm}$ more power to meet the minimum rate requirements of two tags regardless of α. With this additional power, however, the proposed NOMA system can achieve a higher rate for the primary user (Fig. 8) while maintaining the rate requirements of both tags (Fig. 9 and Fig. 10). Additionally, as α decreases, transmit power increases for specific tag rate requirements. This is because the tags reflect less power and harvest more.

In the perfect CSI scenario, Fig. 8, Fig. 9, and Fig. 10 illustrate the primary user rate and tag rates relative to the minimum tag rate threshold. As a benchmark, the tag rates are shown when the BS adopts random beamforming. However, this beamforming lacks the guarantee to meet the primary and tag rate requirements, rendering it unsuitable for SR deployments. In contrast, the proposed beamforming design ensures the fulfillment of these requirements and enables green SR even with low transmit powers.

In Fig. 8, when the tags require zero rates, the goal is only

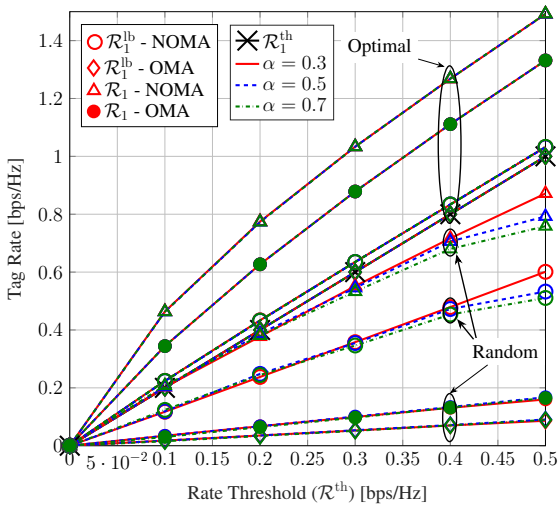

FIGURE 9: T_1 rate as a function of rate threshold ($\mathcal{R}_{\rm th}$).

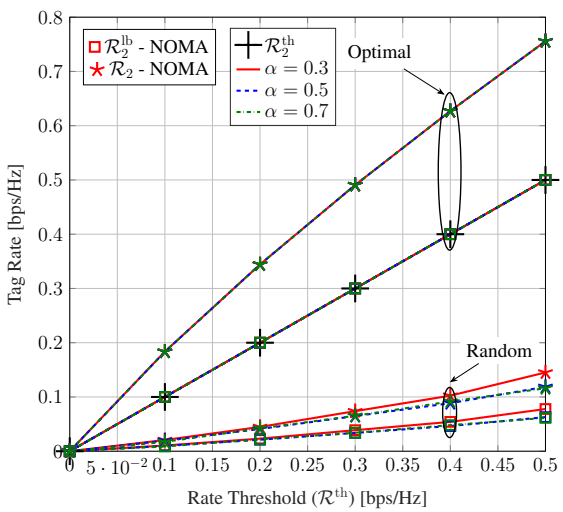

FIGURE 10: T_2 rate as a function of rate threshold ($\mathcal{R}_{\rm th}$).

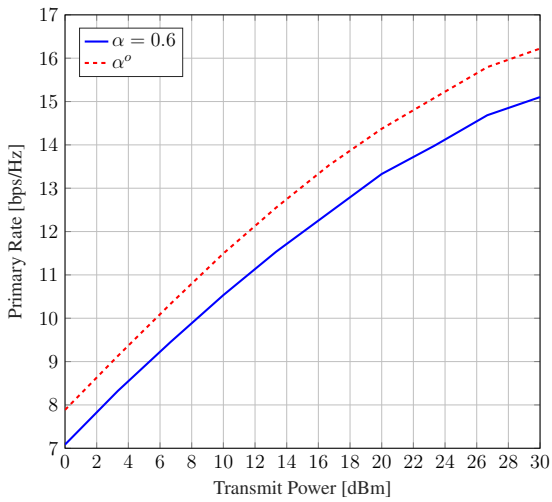

FIGURE 11: The rate of the primary user.

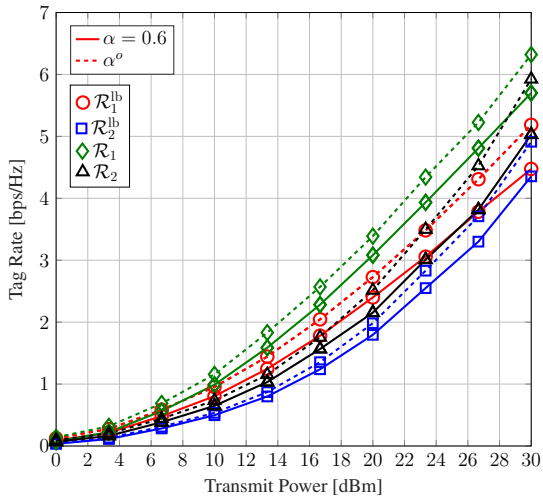

FIGURE 12: The rates of tags.

to maintain the primary user's rate requirement. The primary user reaches higher rates than the respective threshold rates when the tag rate threshold grows. Since the BS allocates more transmit power to maintain the rate requirements of two tags, the rate improvement of the primary user of the proposed NOMA scheme is more significant than that of the OMA scenario. For instance, for $\mathcal{R}_{\rm th} = 0.3\,\text{bps/Hz}$ and $\alpha = 0.5$, this rate improvement is $2\,\text{bps/Hz}$, whereas the primary user of the proposed scheme achieves an $8\,\text{bps/Hz}$ higher rate than the respective threshold rate, $\mathcal{R}_0^{\rm th} = 4\,\text{bps/Hz}$. As per Fig. 8, our proposed transmit power minimization approach ensures the primary user's rate requirement regardless of the tag's α. The user achieves higher rates for low α values than for high α values, owing to higher transmit power being assigned at the BS for low α values, as observed in Fig. 7.

Furthermore, Fig. 9 and Fig. 10 illustrate the tag's actual (12) and lower bound (17) rates of T_1 and T_2, respectively. It is observed that the tight rate lower bounds ($\mathcal{R}_1^{\rm lb}$ and $\mathcal{R}_2^{\rm lb}$) adopted for the proposed transmit power minimization problem satisfied the minimum rate requirements of the tags for different values of α, whereas the actual rate of the tags ($\mathcal{R}_1$ and $\mathcal{R}_2$) is always greater than the required rate. Specifically, since T_2 has the worst channel gain in the proposed NOMA scheme, its lower bound matches the required rate (Fig. 10), while T_1's lower bound is slightly higher (Fig. 9). This is because the tag with the lowest channel gain, i.e., T_2, requires more power to maintain the rate. When the lowest tag's rate is satisfied, the primary user and the rest of the tags, i.e., with higher channel gains, achieve higher rates than their threshold.

B. RECONFIGURABLE TAGS

In the following, we investigate the performance gains of reconfigurable tags whose variable reflection coefficients are optimized through our proposed algorithm, i.e., $\boldsymbol{\alpha} = \boldsymbol{\alpha}^o$ (Section V-A). We thus plot the achieved rates of the user and tags in Fig. 11 and Fig. 12, respectively. Passive tags

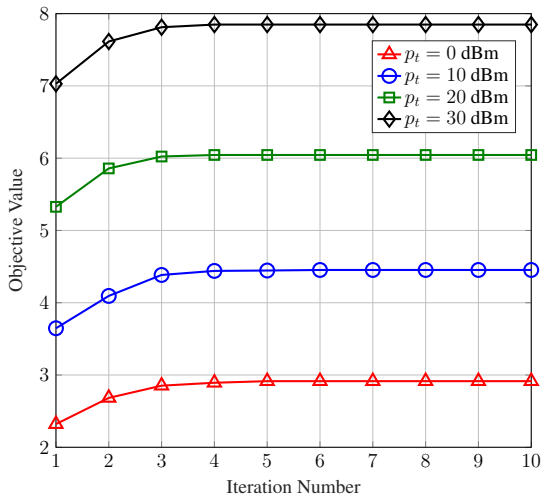

FIGURE 13: The convergence of the objective value of the overall algorithm.

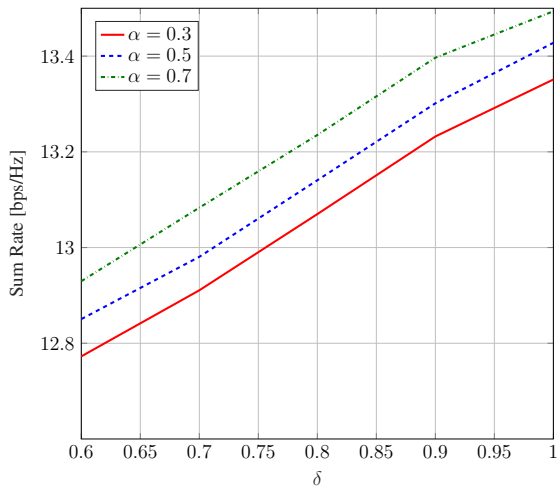

FIGURE 14: The sum rate as a function of imperfect CSI (δ) for $p_t = 20\,\mathrm{dBm}$.

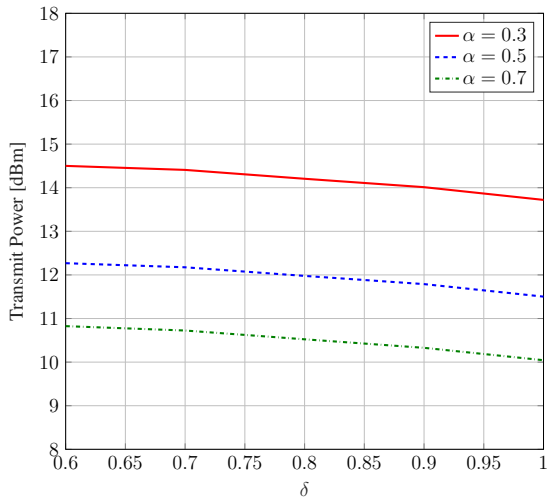

FIGURE 15: The minimum transmit power as a function of imperfect CSI (δ).

with fixed reflection coefficients ($\alpha_k = \alpha = 0.6$,) are also considered as benchmarks. As expected, when the tags use optimal $\boldsymbol{\alpha}^o$, the data rates of both the user and the tags significantly improve. For example, the total sum rate ($\sim$1.8 bps/Hz at $p_t = 20\,\mathrm{dBm}$) exceeds than that of the fixed tags.

In Fig. 13, we explore the convergence rate of our proposed algorithm for reconfigurable tags, as discussed in Section V-A. As our algorithm uses the AO paradigm to break up the non-convex problem, convergence issues are important. However, analytical treatment of such issues appears intractable, but a numerical investigation can yield useful insights. Therefore, the plot depicts the convergence of the objective function, representing the weighted sum rate, as a function of the overall iteration number. We consider four different BS transmit power levels: $p_t = 0, 10, 20, 30, \mathrm{dBm}$. To ensure convergence, we set a stopping criterion that requires the increment of the normalized objective function to be less than $\epsilon = 10^{-3}$. From the observations in Fig. 13, it becomes evident that the weighted sum rate obtained by our algorithm experiences a rapid increase initially and eventually saturates as the number of iterations progresses. An interesting finding is that regardless of the AP transmit power, the algorithm consistently achieves convergence in less than four iterations.

C. ROBUSTNESS AGAINST CSI ERRORS

In Fig. 14 and Fig. 15, we investigate the adverse effects of CSI error to get insight into the interacting mechanism of CSI error and performance degradation of the SR network. Here, we consider two passive tags with fixed reflection coefficients.

Fig. 14 illustrates the sum rate of the user and two tags as a function of the imperfect CSI coefficient δ, considering various reflection coefficients α. Notably, the sum rate improves with better CSI estimation, and as δ approaches 1 (perfect CSI), the rates increase accordingly. For $\alpha = 0.5$ and $\delta = 1$, the rate reaches $13.43\,\mathrm{bps/Hz}$, while with $\delta = 0.7$ (imperfect CSI), the sum rate decreases by $0.44\,\mathrm{bps/Hz}$. Interestingly, the sum rate degradation as δ decreases remains consistent regardless of the value of α. These findings indicate that our proposed algorithms demonstrate a commendable level of robustness against CSI errors.

Fig. 15 also plots the minimum transmit power as a function of the imperfect CSI coefficient (δ) for several α values. This figure shows that for low values of δ, the BS should allocate additional power to counteract the adverse effects of CSI errors while maintaining the primary user's and tags' minimum rate requirements. When δ approaches 1 (perfect CSI), less power is required to maintain the devices' rate requirements. However, the power difference between $\delta = 0.6$ and $\delta = 1$ is as low as $\sim 0.8\,\mathrm{dBm}$ regardless of α, which eventually shows the robustness of the proposed transmit power minimization scheme against CSI errors.

TABLE 3: Maximum tag activation distance.

α	0.1	0.3	0.5	0.7	0.9
$d_{\max}$	$\sim 13\,\mathrm{m}$	$\sim 18\,\mathrm{m}$	$\sim 27\,\mathrm{m}$	$\sim 20\,\mathrm{m}$	$\sim 8\,\mathrm{m}$

D. PASSIVE TAG COVERAGE

The coverage distance for passive tags achieved with the proposed algorithm is a critical factor. As our system model includes path loss, there is indeed a maximum distance of the tags from the BS, i.e., d_{h_k} for $k \in \mathcal{K}$. Clearly, it is desirable to increase the maximum distances. However, if the distances are too large, our algorithms may fail to converge. The convergence of our proposed algorithms is guaranteed within a coverage region depending on the BS transmit power, the number of BS antennas, and path losses. For instance, the coverage distance can be extended by increasing the transmit power of the BS or the number of BS antennas.

Table 3 presents valuable insights into the maximum activation distance ($d_{\max}$) achieved by our algorithms when employing a transmit power of $20\,\mathrm{dBm}$ and a BS equipped with 32 antennas. To obtain these results, we conducted simulations by systematically varying the distances between the BS and users (d_f) and between the BS and tags (d_{h_k}) until the algorithm failed to converge for different values of α.

At low α values, a key limitation arises due to the required rate of the tags, as the reflected power becomes minimal. Conversely, in the moderate α regime, a high activation distance can be achieved by efficiently allocating power for both data transmission and EH. However, when α is high, the algorithm's support is constrained due to the low power allocation for EH, resulting in a limited distance of activation. Table 3 yields valuable insights into the trade-offs between power allocation, activation distance, and the impact of α on the system's performance. These findings shed light on the effectiveness and limitations of our proposed algorithms under different scenarios.

VII. CONCLUSION

In this article, we explore a secondary backscatter network with a primary cellular user and propose an optimal beamforming strategy to serve all devices concurrently. This network is an emerging example of an SR network. However, the resource allocation problems associated with this setup to maximize either the weighted sum rate of the primary user and the tags or the transmit power of the BS while maintaining the minimum rates of the user and the tags and the EH requirements of the tags, are non-convex. Such problems are not amenable to popular convex algorithms. To address this issue, we employ AO, FP, and SDR techniques to determine solutions.

It is important to note that our study includes both fixed and reconfigurable passive tags. The former is economical and provides a feasible networking solution due to its thin, small, and inexpensive nature. Meanwhile, the latter is more expensive but can improve overall network performance. Further research is necessary to evaluate the suitability of each type for various use cases.

Our work also investigates the impact of CSI errors on the proposed optimization schemes and demonstrates their robustness, which is critically important because CSI errors are unavoidable in practice. Another advantage of our solutions is that they allow for multiple access in backscatter systems without modifications or additional processing at the tag while also ensuring that the minimum rate requirements of the primary user and tags are met. As a result, the cost and size benefits of passive tags are preserved, paving the way for the development of ambient IoT solutions.

Overall, our work provides a framework for designing efficient and reliable backscatter systems with multiple users and tags. Furthermore, it highlights the potential of ambient IoT solutions in this context.

While our investigation reveals the broad trends of the symbiotic benefits, here are some specifics of the findings.

1) The proposed optimal beamforming delivers more power to the tags, increasing their EH and rate (Fig. 4). In particular, T_1 and T_2 can reap $4.76\,\mathrm{mW}$ and $3.63\,\mathrm{mW}$, respectively, with 32 antennas and $20\,\mathrm{dBm}$ transmit power at the BS, and $1.56\,\mathrm{bps/Hz}$ and $1.23\,\mathrm{bps/Hz}$.
2) When determining the appropriate reflection coefficient, it is crucial to strike a balance between EH and rate. This delicate performance trade-off is visualized by the trade-off curve shown in Fig. 4. Moreover, the reflection coefficient α choice significantly impacts the performance of the primary user. Specifically, it influences the interference caused by the tags to the primary user's operation, as illustrated in Fig. 8. Hence, selecting an optimal α value is of utmost importance to ensure an efficient and interference-free operation for the primary user.
3) The optimal BS beamforming vector for weighted sum rate maximization can enhance the performance of both the primary user and the tags while meeting rate and EH requirements. For example, with $20\,\mathrm{dBm}$ transmit power, 32 antennas, and $\alpha = 0.6$, the user, T_1, and T_2 rates are $13.4\,\mathrm{bps/Hz}$, $2.8\,\mathrm{bps/Hz}$, and $2\,\mathrm{bps/Hz}$, respectively (Fig. 5 and Fig. 6).
4) The proposed transmit power minimization reduces the energy use of the BS while preserving primary user and tag rates (Fig. 7, Fig. 8, Fig. 9, and Fig. 10). With $\alpha = 0.5$, the user, T_1, and T_2 achieve a minimum rate of $12.0\,\mathrm{bps/Hz}$, $1.0\,\mathrm{bps/Hz}$, and $0.5\,\mathrm{bps/Hz}$, respectively, whereas the BS needs just $11.7\,\mathrm{dBm}$.
5) The proposed weighted sum rate maximization and transmit power minimization algorithms (Algorithm 1 and Algorithm 2, respectively) are resistant to CSI errors (Fig. 14 and Fig. 15). In particular, the sum rate and transmit power differences between the worst (i.e., $\delta = 0.6$) and perfect CSI cases are respectively $\sim 0.58\,\mathrm{bps/Hz}$ and $\sim 0.8\,\mathrm{dBm}$, regardless of α.

6) Optimal reflection coefficients of the reconfigurable tags can be chosen to significantly improve the sum rate, i.e., $\sim 1.8\,\mathrm{bps/Hz}$ sum rate improvement at $p_t = 20\,\mathrm{dBm}$ (Fig. 11 and Fig. 12). Of course, such tags are more expensive and power-consuming than passive ones.

While we comprehensively investigated a symbiotic network (Fig. 2), further research is necessary to address additional challenges. In particular, when multiple primary users are present, interference terms need to be taken into account, which can affect the effectiveness of resource allocation algorithms. Moreover, developing user and tag clustering approaches is also important to improve the overall system performance. Although the proposed solutions perform well in the considered simulation scenarios, coverage of passive tags over long distances can be problematic (Table 3). Due to severe path losses, our algorithms may fail to activate the tags while satisfying EH and data rate targets. Therefore, future research efforts should focus on addressing these challenges to develop the SR concept.

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